

Multi-Disease Detection System Using Yolo V8 in Apricot Fruit



Session: 2020-2024

Submitted By

Saqib Hussain

FA20-BSE-031

Wazir Muhammad Muzammil

FA20-BSE-038

Ahsan Azam

FA20-BSE-044

Supervised by
Engr. Aqsa Rasheed

Department of Software Engineering
Mirpur University of Science and Technology, Mirpur
Spring 2024

A Dissertation Submitted To
Department of Software Engineering, Mirpur
University of Science and Technology, Mirpur
AJK As a Partial Fulfillment of the Requirement for the
Award of the
Degree of
Bachelor of Software Engineering

Final Approval

This is to certify that we have read the report submitted by **Saqib Hussain (FA20-BSE-031)**, **Wazir Muhammad Muzammil (FA20-BSE-038)**, **Ahsan Azam (FA20-BSE-44)** for the partial fulfillment of the requirements for the degree of the Bachelors in Software Engineering (BSE). It is our judgment that this report is of sufficient standard to warrant its acceptance by Mirpur University of Science and Technology for the degree of the Bachelors in Software Engineering (BSE).

Committee:

1

Ms. Aqsa Rasheed
(Supervisor)

2

[Name of External Examiner]
(External Examiner)
Date: [date of external examination]

3

Dr. Nouman Ali
(Head of Department/chairman)

UNDERTAKING

We hereby declare that this document “**Multi-Disease Detection System Using Yolo V8 in Apricot Fruit**” neither as a whole nor as a part has been copied out from any source. It is further declared that we have done this project with the accompanied report entirely on the basis of our personal efforts, under the proficient guidance of our teachers especially our supervisor **Ms. Aqsa Rasheed**. If any part of the system is proved to be copied out from any source or found to be reproduction of any project from anywhere else, we shall stand by the consequences.

Saqib Hussain

FA20-BSE-031

Wazir Muhammad Muzammil

FA20-BSE-038

Ahsan Azam

FA20-BSE-044

LIST OF FIGURES

Figure 2.1 Map of study	6
Figure 3.1 Methodology.....	15
Figure 3.2 Dataset Collection.....	15
Figure 3.3 Original and Annotated images for apricot dataset	16
Figure 3.4 Block diagram for apricot fruit disease detection system.....	17
Figure 3.5 Architecture of YOLO algorithm	18
Figure 3.6 Workflow of apricot disease detection	22
Figure 3.7 yolov8 predicting sample as healthy.....	22
Figure 3.8 Yolov8 predicting sample as Infected	23
Figure 3.9 Yolov8 predicting sample as infected.....	23
Figure 4.1 Samples of data.....	27
Figure 4.2 Confusion Matrix.....	29
Figure 4.3 F1curve of yolo v9.....	31
Figure 4.4 Confusion matrix of yolo v9.....	32
Figure 4.5 Precision-curve for Yolo v9.....	32
Figure 4.6 Recall curve of yolo v9.....	33
Figure 4.7 Training losses	33
Figure 4.8 Confusion Matrix Yolov08.....	34
Figure 4.9 Confusion Matrix Normalized Yolov08	34
Figure 4.10 F1-Confidence Curve.....	35
Figure 4.11 Precision of yolo v8.....	35
Figure 4.12 Recall of yolo v8.....	36
Figure 4.13 Training losses	36
Figure 4.14 F1-Confidence Curve Yolov5.....	37
Figure 4.15 Reacll Curve Yolov5	37
Figure 4.16 Precision curve Yolov5.....	38
Figure 4.17 Training losses in Yolov5.....	38
Figure 4.18 CNN Model Accuracy	39
Figure 4.19 CNN Model Loss.....	40
Figure 4.20 Model Accuracy	41
Figure 4.21 VGG Model Loss.....	42
Figure 4.22 YoloNAS Precision	42
Figure 4.23YoloNAS Recall	43
Figure 4.24 Traning Loss	43

Table of Content

Content	Page No
Final Approval	3
UNDERTAKING	4
LIST OF FIGURES	5
IST OF TABLES	8
ABBREVIATIONS	9
ACKNOWLEDGEMENTS.....	10
ABSTRACT.....	11
1. Introduction	2
1.1 Goals.....	2
1.2 Motivation	2
1.3 Problem Statement.....	3
1.4 Method	3
1.5 Report Overview	3
1.6 Conclusion.....	4
2. Literature Review	6
2.1 Classification.....	7
2.2 Detection Using Deep Learning Techniques	8
2.3 Segmentation by Artificial Neural Networks	9
2.4 Detection Model Using Machine LearningTechniques	9
2.5 Research Gaps	10
2.6 Research Questions	11
2.7 Semantic Literature Review	11
2.8 Conclusion.....	12
3. Proposed Methodology	14
3.1 Introduction to proposed methodology	14
3.2 Proposed Methodology.....	15
3.3 Dataset.....	15
3.4 Data Preprocessing	16
3.5 Data Acquisition.....	16
3.6 Block diagram and explanation.....	16
3.7 Deep Learning Model	17

3.8	Yolo Architecture	18
3.9	YOLOv8.....	19
3.9.1	Backbone	19
3.9.2	Neck.....	19
3.9.3	Head.....	20
3.10	Training	20
3.11	Hardware specifications	21
3.12	Disease prediction	21
3.13	Disease prediction via Yolov8	22
3.14	Conclusion.....	24
4.	Implementation.....	26
4.1	Data Collection.....	26
4.2	Procedures	27
4.3	Performance Matrix.....	27
4.3.1	Precision	27
4.3.2	Recall or Sensitivity	28
4.3.3	Mean Average Precision (mAP).....	28
4.3.4	Confusion Matrix	28
4.4	Results and Discussion.....	29
4.5	State-of-the-art comparison.....	30
4.6	Graphs	31
4.6.1	Yolo v9	31
4.6.2	Yolo v8.....	34
4.6.1	Yolo v5	37
4.6.2	CNN.....	39
4.6.1	Vgg16	41
4.6.2	Yolo Nas.....	42
4.7	Conclusion.....	43
5.	Conclusion.....	46
6.	Limitation	46
7.	Future work	47
8.	Reference.....	49

LIST OF TABLES

Table 2.1 Semantic Literature Review	12
Table 3.1 Details of yolov8 hyperparameters.....	21
Table 4.1 Result and discussion	30
Table 4.2 State-of-the-art comparison among models.....	31

ABBREVIATIONS

FYP:	Final year project
ML:	Machine learning
DL:	Deep learning
ROC:	Receiver Operating characteristics
SVM:	Support Vector Machine
YOLO:	You only look once
CNN:	Convolutional neural network
CSRE:	Class specific recognition error
ELM:	Extreme learning machine
ANN:	Artificial neural network
mAP:	Mean Accuracy Precision
SNN:	Simulated neural network

ACKNOWLEDGEMENTS

First of all, we are obliged to Allah Almighty the Merciful, the Beneficent and the source of all Knowledge, for granting us the courage and knowledge to complete this Project. We can never thank Him enough for His countless blessings upon us. Praise to Prophet Muhammad (S.A.W), who is and will always be a source of guidance and knowledge for humanity as a whole. I am thankful to all those persons because the activity if going through industrial orientation it'll bridged me to become good IT Professional. Naturally, it requires lot of people support to complete this project. We take this opportunity to acknowledge their support to us. First of all, I am extremely thankful to Mirpur University of Science, Software Engineering department and for giving permission to do project. We consider our self-privileged having found an opportunity to express our heartily thanks to our Supervisor Ms. Aqsa Rasheed. The success of any project is never limited to the individual undertaking the project. However, there are some key personalities whose role is very vital in project. So, I greatly obliged to express my gratitude to all the mentioned officials of MUST Software Engineering Faculty for giving me assistance, guidance, inspiration in this project.

Saqib Hussain

FA20-BSE-031

Wazir Muhammad Muzammil

FA20-BSE-038

Ahsan Azam

FA18-BSE-044

ABSTRACT

Fruits are essential and nutrient-rich cash crops globally, with apricots being particularly significant. However, the cultivation of apricots is challenged by various diseases that can adversely affect fruit yield and quality. Traditional manual methods for disease identification in large volumes of apricots are labour-intensive and inefficient. Our project addresses this challenge by proposing an advanced approach for detecting multiple diseases affecting apricot fruits. To improve the accuracy and efficiency of disease detection, we introduce a novel methodology that utilizes sophisticated detection techniques. Object detection can be used to identify and track objects and people, which can help for automatic disease detection of apricot fruit. Firstly, we have created dataset then secondly, then images are manually annotated, which is a process where a human expert goes through each image and marks the location and class of objects within the image. This process is important for training object detection algorithms as it provides the necessary ground truth data for the algorithm to learn from. Lastly, a comparative analysis has been performed and achieved a state-of-art result using YOLOv8. This approach aims to identify and classify three specific diseases impacting apricot fruits. The performance metrics we have attained are Precision of 92.5%, Recall of 92.2%, and a mean Average Precision (mAP) of 95%. These metrics highlight the effectiveness of the proposed method in accurately detecting and classifying diseases. In addition to evaluating the model, this research provides a detailed analysis of various detection techniques and visualization methods used in plant disease management. It also identifies research gaps and suggests future directions to enhance disease detection capabilities, aiming for earlier identification even before visible symptoms appear. By advancing the techniques for apricot fruit disease detection, this thesis not only contributes to more efficient management practices but also plays a crucial role in enhancing food security. To achieve this, we collected apricot fruits directly from the field and developed a comprehensive dataset. This dataset, which includes diverse samples affected by various diseases, serves as a foundation for training and validating our detection methods. By leveraging this real-world data, our approach supports more accurate disease identification and classification, ultimately improving the sustainability and productivity of apricot cultivation. This work ensures that apricot farmers have access to reliable tools for early disease detection, thereby safeguarding their crops and contributing to broader food security efforts.

Chapter 1

INTRODUCTION

CHAPTER 1:

1. Introduction

Among the most nutrient-dense and commercially important cash crops in the world are fruits, of which apricots are a notable example. But as time goes on, these fruits become more susceptible to a number of illnesses, like brown rot and shot hole, which can seriously reduce both yield and quality. In addition to being labour-intensive and time-consuming, the manual method of diagnosing these diseases in big quantities of fruit requires a major human resource investment. For efficient disease control and the preservation of crop health, early identification and accurate diagnosis of these illnesses are essential.

We suggest a thorough multi-disease identification technique designed especially for apricot fruits to overcome these issues. Many Machine Learning (ML) models have been used for plant disease identification and categorization; however, new developments in Deep Learning (DL) have demonstrated impressive potential for improving precision in this field. In order to identify and categorize the signs of apricot fruit disorders, this study explores a variety of created and altered deep learning architectures in addition to visualization methods. It also uses a number of performance measures to assess how well these methods work. In addition, this work points out important areas for future research with the goal of enhancing the transparency and precision of disease detection in apricot fruits before symptoms manifest.

1.1 Goals

Our goal is to create a reliable and effective technology that can identify a variety of apricot fruit illnesses with accuracy. If this initiative is carried out successfully, it may completely change the way apricot illnesses are treated, which would greatly advance the long-term viability of apricot orchards. Through the use of sophisticated machine learning algorithms and cutting-edge deep learning techniques, this device will carefully examine digital photos of apricots in order to determine the kind and severity of illness that are present.

Getting practical expertise in web and app development is another crucial goal. Our goal in deciding to use both app and online platforms for our Final Year Project (FYP) is to get a variety of abilities that will allow us to operate effectively on both after the project is finished. By using a dual platform approach, we are able to expand our software development knowledge in addition to improving the project's practical usefulness.

1.2 Motivation

The urgent need to address issues in the agricultural sector, particularly in fruit production, is the driving force for this study on "Multi-Disease Detection using Yolov8 in Apricot Fruit." Diseases that can significantly lower productivity and quality are becoming a bigger hazard to apricot agriculture. This project aims to revolutionize apricot farming operations by expanding the use of machine learning algorithms for early disease diagnosis and offering a dependable,

effective, and non-invasive approach.

This research has the potential to have a significant impact since it advances crop management techniques and promotes sustainable farming methods, which will eventually increase yields, decrease losses, and improve food security. This initiative offers a promising way to increase the resilience and productivity of apricot farming, marking a significant advance at the nexus of agriculture and technology.

1.3 Problem Statement

The current problem with the traditional method is Subjectivity and inaccuracy of visual inspection, Time-consuming and labor-intensive process and Limited accessibility for early detection.

1.4 Method

A number of critical steps are included in the machine learning method for multi-disease identification in apricot fruit to ensure accurate and reliable diagnosis of various ailments. Taking photos of apricot fruits with shot holes and brown rot is the first step in the process. Subsequently, the image data undergoes a preprocessing step to enhance the image quality and remove any noise or unnecessary information. Preprocessing is necessary to optimise the data for feature extraction and analysis later on.

After preprocessing, the apricot fruit pictures are examined using sophisticated feature extraction algorithms to identify distinctive patterns and illness symptoms. These collected features serve as the foundation for training machine learning algorithms, which enables the classification and categorisation of the many diseases affecting the fruit samples. The machine learning model needs to be correctly trained and relevant characteristics need to be selected with attention in order to achieve high sickness detection accuracy.

Once the machine learning model has been trained, it undergoes testing and validation using a separate set of apricot fruit images with known disease conditions. This validation phase is crucial to ensure that the model can accurately generalize its detection capabilities to new and unseen fruit samples. Following validation, the overall performance of the multi-disease detection system is rigorously evaluated, and any necessary refinements or optimizations are implemented to enhance both accuracy and efficiency.

IT create a reliable, automated system for accurate and timely disease identification, the machine learning approach for multi-disease detection in apricot fruit combines advanced image processing, feature extraction, and machine learning algorithms. This novel method has the potential to completely transform the way apricot fruit diseases are managed, which would ultimately help apricot orchards be grown in a sustainable manner.

1.5 Report Overview

This report introduces a system designed to provide readers with an understanding

of the foundational concepts. The report is organized into five chapters, each focusing on different aspects of the system.

Chapter 1 presents an overview of the system, including its introduction and primary objectives. It outlines the key factors involved in the development process and provides a summary of the system's core ideas and concepts.

Chapter 2 delves into the background of the project, identifying the problem statements that prompted the system's development. This chapter also includes a comprehensive literature review, covering critical success factors in project management and user interaction. The review discusses project success factors, critical success factors, and the role of social interaction among users.

Chapter 3 details the methodology employed in developing the project. It describes the methods and approaches used, supported by detailed diagrams that illustrate the project's structure and process.

Chapter 4 focuses on the implementation phase, where multiple deep learning models are applied to determine which model is most effective for detecting diseases in apricot fruits.

Chapter 5 provides an in-depth discussion of the project's findings and outcomes, offering a comprehensive overview of the project and summarizing the conclusions drawn from the research and implementation stages.

1.6 Conclusion

In conclusion, this study presents a robust framework for the detection of multiple diseases in apricot fruits using advanced deep learning techniques. We greatly minimize the time, labor, and subjectivity associated with existing methods by automating the disease identification process. The application of advanced machine learning algorithms with Yolov8 has shown great promise for precise illness categorization, improving the capacity to identify early warning indicators of fruit problems. When properly put into place, this approach can reduce crop losses and increase apricot orchard productivity, which in turn can support sustainable farming practices. Furthermore, the study demonstrates the revolutionary potential of incorporating technology into agriculture and establishes the foundation for future developments in plant disease identification. The dual-platform approach further underscores the practical applicability and flexibility of the system, preparing it for real-world deployment on both web and mobile platforms.

Chapter 2

LITERATURE REVIEW

CHAPTER 2:

2. Literature Review

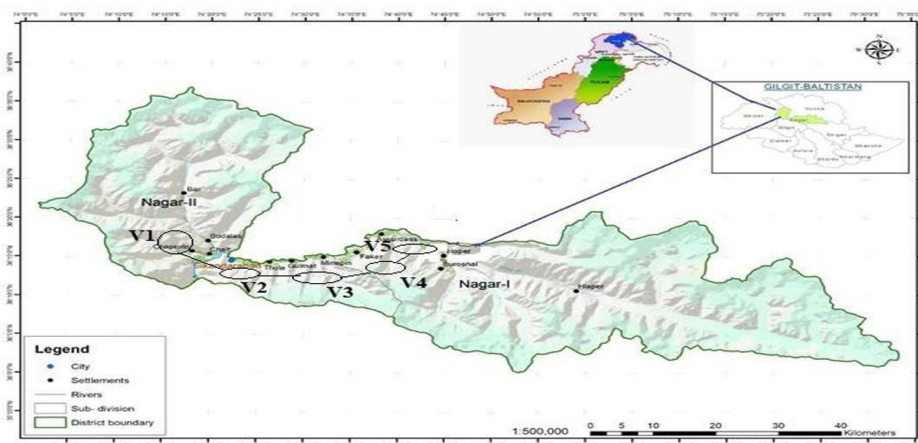
Although a lot of previous research has been done in the field of plant diseases, our work provides a thorough analysis of apricot shot hole disease, which poses a significant risk to apricot production in Pakistan's Gilgit-Baltistan region. This study offers a thorough examination of the disease's prevalence, incidence, severity index, and geographic distribution over several orchards and valleys using descriptive statistics and geostatistical approaches.

The results show that there is a considerable geographical range in the prevalence of the condition, with rates ranging from 50.0% to 67.70%. Further evidence that environmental factors or other spatial variables have a substantial impact on the disease's distribution comes from geostatistical analysis, which shows a moderate spatial dependence for incidence, severity index, and overall disease severity.

Our Project adds significantly to our understanding of apricot shot hole , Brown rot diseases by providing important new information that can inform preventative and therapeutic measures. The findings are especially useful for creating focused treatments to lessen the disease's effect on apricot output in the area, including precision fungicide sprays or certain cultural practices. Additionally, the study emphasizes the significance of taking spatial aspects into account when evaluating and managing plant diseases, emphasizing the need for site-specific strategies.

This work advances knowledge of the dynamics of plant diseases and their geographic distribution, which has consequences for disease management that go beyond its immediate use. The knowledge acquired can be crucial in creating early warning systems and predicting models for disease outbreaks, allowing for preemptive steps to protect agricultural crops from possible dangers.

In conclusion, this comprehensive investigation of apricot shot hole disease in Gilgit-Baltistan provides a valuable contribution to the field of plant pathology and agricultural disease management. The findings offer insights into the disease's spatial dynamics and inform the development of effective strategies to mitigate its impact on apricot production in the region. Figure 2.1 show the map of district Nagar apricot production.



M. Muhammad et al,[1] Apricot (*Prunus armeniaca*. L) is the most popular fruit of Gilgit-Baltistan (GB) and is grown in all ten districts of the area. Agriculture Statistics, 2017 states that 107,737 tons of apricots are produced annually from more than 2.72 million apricot trees spread across 6,370 227 hectares. Due to its large agricultural revenue share, it has a big impact on the socioeconomic development of the area. 480 apricot trees, 30 orchards, and 5 valleys make up the geographical distribution of the apricot shot hole disease in the districts of Gilgit-Baltistan, Pakistan, which was assessed using geostatistical methods. To determine the incidence, illness index, and severity, 150 apricot fruits, or a total of 72,000, were randomly chosen from each apricot tree. The disease was shown to be differently distributed in space throughout the research area.

2.1 Classification

According to C. Chen et al. [2], managing resources becomes more difficult with the standard variety identification method because it takes longer. In order to reliably categorize apricot fruits, a study created a dataset and a U-net model that allowed researchers to swiftly detect variety information. To handle apricot scientific data effectively, a variety search engine, mobile/web application (ApricotView), and apricot database platform (ApricotDIAP) were developed. These tools have a high degree of accuracy when it comes to recognizing apricot varieties using image recognition models. With a high degree of classification accuracy, the U-net model is used to finely segment apricot fruits according to their size, shape, and color properties. Through transfer learning, it expedites training and achieves an F-score of 99% for fruit image segmentation. With the help of the model, a variety search engine that can

Mehmood, A. et al., [3] The study focusses on automated fruit disease detection and precision farming. The amount of data, the variety of fruit varieties, and the types of diseases present in the literature are constraints. With 100% accuracy for fruit classification and 99% accuracy for illness identification, the suggested ensemble classification method surpasses alternative approaches. The ROC curve study demonstrated that the ensemble classification approach performed significantly better than the other classification methods. The study addressed the requirement for a more extensive dataset with a variety of fruit types and diseases. The goal of the study is to improve fruit disease classification and detection in precision agriculture, which is essential for ensuring the world's food security.

Amari, K. and others, 2007 [4] This study investigates the effects of viroids on Spanish apricot trees, focusing specifically on the Murcia region. The predominant ailment identified was Hop Stunt Viroid (HSVd), associated with the fruit condition known as "degeneration." Fruit infected with HSVd loses some of its ideal outward look (organoleptic features) along with its desired taste and texture. Studies confirm the presence of HSVd in symptomatic apricot trees, with higher frequency in branches bearing symptomatic fruits. This strong link indicates the significant contribution viroids make to the onset of apricot diseases in Spain, which result in unmarketable fruit and significant financial losses for growers.

M.H. Saleem et al., [5] To maintain healthy growth, it is imperative to detect plant illnesses early on, and Deep Learning (DL) models are becoming effective tools for this purpose. Scholars have examined diverse deep learning architectures and visualization methodologies. Their findings indicate that cascaded VGG and inception modules outperform well-established models like AlexNet and Inception-v3 in terms of testing accuracy. A major obstacle still needs to be overcome, though: obtaining transparency in illness identification prior to the onset of obvious symptoms. This draws attention to a critical gap in the literature and indicates that there is room for innovation in plant disease detection if DL models are enhanced to classify diseases across their whole life cycle.

2.2 Detection Using Deep Learning Techniques

Chowdhury, M. E. et al., [6] The automatic identification of leaf disease using deep learning techniques is the main emphasis of this work. The study classifies tomato pathogens more accurately than previous research by using the EfficientNet deep learning architecture. Using segmented leaf pictures, the study produced high classification accuracies for binary, six-class, and ten-class classification. Specific gaps in the body of current literature are not mentioned in the study. The potential for automated plant disease monitoring was demonstrated by the successful classification of tomato illnesses with the application of deep learning and segmentation models such as U-net and EfficientNet.

M. TÜRKOĞLU, [7] This study explores the use of deep learning features for plant disease diagnosis. While a number of well-known deep learning networks, such as AlexNet and VGG models, have been investigated for use in transfer learning to identify pests and diseases in agriculture, this study highlights the efficacy of deep feature extraction in conjunction with SVM or ELM classifiers. When compared to transfer learning alone, their data shows that this technique produces better results. There is still a need to investigate a larger variety of deep learning architectures or possible pairings for additional performance gains. All things considered, the study emphasizes how important deep feature extraction and particular model layers are to getting precise plant disease diagnosis.

In 2020, Chohan, M. et al. [8] To detect diseases in plant leaves, scientists are using a deep learning model called Plant Disease Detector. In order to increase accuracy and sample size, this model makes use of a multi-layered convolutional neural network that was trained on the Plant Village dataset. Although the system achieved a high testing accuracy of 98.3% on the dataset, its true potential can be realized by adding photos of real-world environments. In the end, this would open the door for integration with drones or agricultural systems for real-time disease identification and better crop health management. It would also improve its capacity to handle a variety of leaf conditions and categories a wider spectrum of plant diseases.

2.3 Segmentation by Artificial Neural Networks

Unay and associates [9] This study employs artificial neural networks to segment defects in bi-colored apple specimens. For the purpose of segmenting surface flaws on apples, researchers have mostly employed Bayesian-based classification techniques and global thresholding-based algorithms. Region-based, classification-based, and unsupervised classification approaches are some of the several segmentation strategies. Because the amount of the defects in the database varies, the research uses class-specific recognition error (CSRE) to assess segmentation performance. The research highlights the need for more precise segmentation techniques by addressing the segmentation of skin flaws on apples using complex classifiers. When compared to competitive networks, the application of artificial neural networks, such as feed-forward and recurrent networks, produced more accurate segmentation results.

2.4 Detection Model Using Machine Learning Techniques

In 2020, Soni, V.D. et al. [10] Diagnosing chronic diseases such as diabetes, lung cancer, and breast cancer can be extremely difficult and time-consuming. Conventional approaches frequently depend on symptoms, which can be arbitrary and restrict useful therapeutic approaches. Through a variety of categorization techniques, including hybrids of genetic algorithms and Support Vector Machines (SVM), machine learning presents a possible option. Studies that have examined the efficacy of various models, such as SVM and ANN, have shown that they have the ability to achieve high classification accuracy for diseases. It is essential to assess these models using metrics such as sensitivity, specificity, and accuracy in order to determine their advantages and disadvantages. One study demonstrated the value of precise prediction in light of symptom-based diagnosis's limitations and the high accuracy that the Sequential Minimal Optimization (SMO) technique can accomplish.

(2018) Badage, A. et al. [11] Plant diseases significantly lower crop yields in agriculture, hence early detection is essential. By using remote sensing photos for automated disease identification, machine learning provides a solution and allows for prompt interventions. Current methods for disease identification include edge detection algorithms (e.g., Canny's), Gaussian filtering, and intensity gradients in image processing. To precisely diagnose diseases, this field of research focusses on spectral features, image processing, and classification algorithms. To accomplish this identification, existing algorithms use training datasets that contain samples of both healthy and ill subjects. Nonetheless, the agricultural industry needs to investigate more sophisticated techniques for routine crop monitoring and disease management measures. Utilizing remote sensing data in conjunction with machine learning, we can greatly increase agricultural output and

U. Fulari et al. [12] Plant phenotyping, which examines a plant's outward features, is an essential instrument for tracking plant development and

identifying illnesses. For this, machine learning methods such as SVM and CNN have proven to be effective tools, with image processing being crucial in the diagnosis of diseases. For infected leaves, Support Vector Machines (SVM) classifiers show a promising accuracy of about 80%; however, Convolutional Neural Networks (CNNs) show even greater disease recognition performance. However, because accuracy varies based on the input data, choosing the best classification technique is still difficult. This demonstrates the necessity of effective and flexible agricultural disease detection technology in order to maximize crop yield and reduce financial losses.

2.5 Research Gaps

To fully realize the potential of deep learning for multi-disease detection in apricot fruit, a number of research gaps must be filled. The scarcity of high-quality, labelled datasets that are specifically designed for apricot diseases is a major obstacle. Large datasets are necessary for deep learning models to be trained effectively, and their limited availability may impair the models' generalizability and performance. Furthermore, it's important to give considerable thought to selecting between multi-class and multi-label categorization techniques. Depending on the approach, the model might have to distinguish between several diseased and healthy states or recognize every illness that is present in a single image. Both strategies have unique benefits and drawbacks, therefore more research is necessary to determine which is the most useful for detecting apricot illness.

The issue of data imbalance, wherein some diseases may be under-represented in the dataset relative to others, is another crucial concern. If this imbalance is not correctly corrected, it may result in a model that gives more weight to common diseases while undervaluing rarer ones, which would lower the system's overall accuracy and dependability. To guarantee thorough illness identification, strategies to reduce this bias, such as data augmentation or resampling, must be investigated.

Furthermore, deep learning models are sometimes viewed as "black boxes," which makes it challenging to comprehend how they make judgements. This lack of clarity may make it difficult for these models to be used in real-world situations. Users' trust and acceptance of the model could be increased by incorporating explainability techniques, which offer insights into the model's decision-making process, especially in agricultural contexts where the stakes are high.

These difficulties offer fascinating chances for additional study and advancement. We may improve the suggested methodology for multi-disease detection in apricot fruit by addressing the issues of data availability, classification strategies, data imbalance, and model explainability. In the end, this will contribute to more efficient and sustainable agricultural operations.

2.6 Research Questions

How does the size and diversity of the training dataset impact the model's accuracy for multi-disease detection?

Can transfer learning from pre-trained models on similar fruit classification tasks be beneficial for our apricot disease detection model?

What metrics are most appropriate for evaluating the performance of a multi-disease detection model for apricot fruit (considering accuracy, precision, recall for each disease class)?

How can the deep learning model be integrated into a user-friendly system for deployment in real-world agricultural settings?

2.7 Semantic Literature Review

As we are going to perform detection of disease through deep learning model so Table 2-1 is brief semantic literature reviews of the previously done projects and research in the domain is here in tabular representation.

Reference	Models	Precision	mAP	Recall	Accuracy	F1-score
[2]	U-Net	-	-	-	-	99%
	Vgg16	-	-	-	97%	-
	KNN	-	-	-	-	-
[3]	CNN	-	-	-	100%	-
[5]	CNN	-	-	-	98.3%	-
[6]	Alex-Net	96.3%	-	96.5%	97.354%	-
	Google-Net	96.989%	-	96.78%	97.711%	-
	SVM	95.4%	-	92.5%	92.9%	-
	Random Forest	94.6%	-	93.8%	93.1%	-
[7]	AlexNet	-	-	-	-	-
	KNN	-	-	-	-	-

	Vgg16	-	-	-	-	94.8%
	Vgg19	-	-	-	-	-
[8]	CNN	96.8%	-	98.6%	98.5%	97.7%
[12]	CNN	-	-	-	98.6%	-

Table 2.1 Semantic Literature Review

2.8 Conclusion

This literature review highlights the significant advancements made in the fields of plant disease detection, classification, and segmentation using machine learning and deep learning techniques. Research has demonstrated the promising potential of several models, including CNN, EfficientNet, and U-net, in accurately classifying fruit illnesses. There is still, however, issues with model transparency, data imbalance, and dataset availability, especially when it comes to the identification of apricot disease. The field could be further advanced by addressing these issues with larger datasets, better classification methods, and increased explainability. By filling in these gaps, our research will aid in the creation of scalable, accurate, and efficient multi-disease detection algorithms for apricot fruits. Significant improvements in agricultural practices and food security will result from this effort, particularly in apricot-producing regions such as Gilgit-Baltistan.

Chapter 3

PROPOSED METHODOLOGY

CHAPTER 3:

3. Proposed Methodology

For a project like disease detection in apricot fruit, an organized strategy that includes important stages like data collection, model building, training, evaluation, and deployment is crucial when presenting a deep learning model technique. This article presents a novel approach that uses deep learning to identify various pathologies in apricot fruit. Deep learning techniques have demonstrated great effectiveness in picture classification tasks, making them especially suitable for automating fruit disease detection. With our method, there is no need for manual inspection, saving time and resources, as deep learning is used to automate the disease identification process. Furthermore, deep learning models are particularly good at deciphering intricate patterns from picture data, leading to more precise disease

This approach has the potential to improve general crop health monitoring procedures in addition to revolutionizing the management of apricot fruit diseases. Farmers can take proactive steps to reduce the spread of disease, enhance treatment plans, and eventually increase the volume and quality of their apricot harvests by automating the diagnosis process.

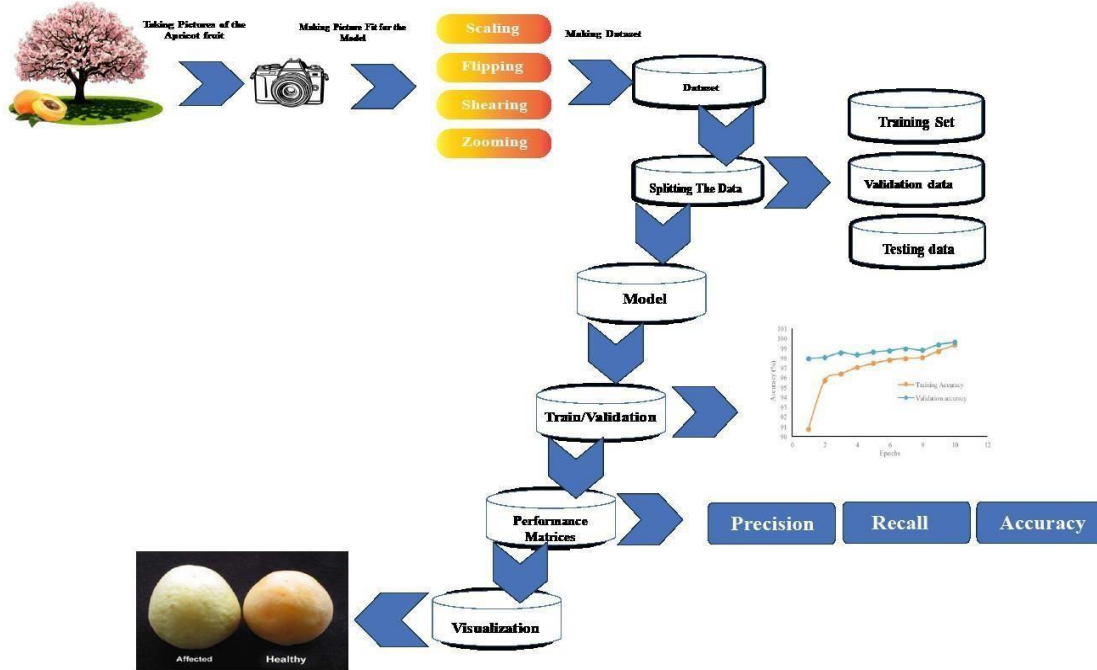
3.1 Introduction to proposed methodology

Deep learning has many important benefits when it comes to illness detection, such as the ability to analyse large volumes of data quickly, adjust to changing disease patterns, and achieve high accuracy. Through the use of a varied dataset that includes photos of both unhealthy and healthy apricot fruits, a model is trained to enable the system to recognize various disease indicators and make precise real-time diagnoses. Utilizing the capabilities of deep learning algorithms, this method looks to identify common diseases like brown rot and shot hole in apricot fruit by analyzing photos. By using cutting-edge deep learning architectures, this system can provide accurate and dependable findings, which makes it an effective weapon in the fight against illnesses that affect fruit crops.

The use of deep learning to illness detection presents a number of compelling benefits, such as the capacity to analyse large volumes of data in an efficient manner, adjust to the complexity of various disease patterns, and attain remarkably high accuracy rates. The technology is able to accurately diagnose patients in real time by successfully learning to discern between various disease symptoms thanks to the model's enormous training dataset of photos of both diseased and healthy apricot fruit. This method uses deep learning algorithms to identify common diseases like brown rot and shot hole by analyzing photos of apricot fruit. With the use of sophisticated deep learning architectures, the system is able to identify and diagnose various medical diseases with a high degree of precision by extracting complex information from these photos.

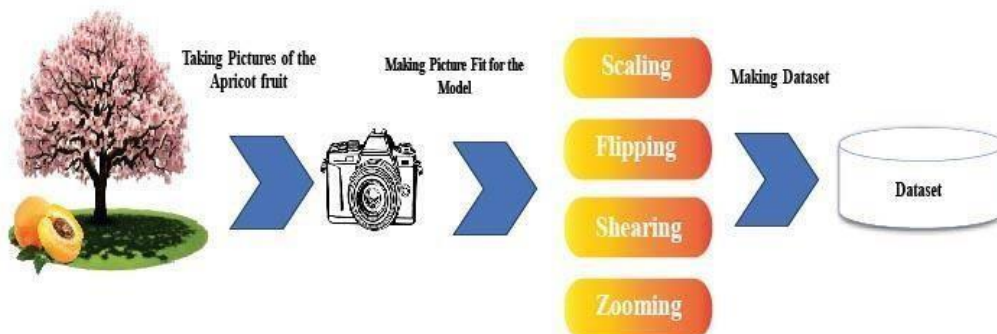
3.2 Proposed Methodology

This Figure 3.1 visualizes the comprehensive approach to training our apricot disease prediction model. It highlights the data collection methods, data preprocessing techniques used to ensure data quality, and the final disease prediction stage.



3.3 Dataset

In order to train a deep learning model required a huge dataset The success of a deep learning model for multi-disease detection hinges on the quality and comprehensiveness of the dataset used for training. The Figure 3.2 shows how wecreate our dataset. Through a website named roboflow.com an online tool we have annotated all the images in order to make the dataset here have a look of the original and annotated



images of infected images and original images. The Figure 3.3 show annotation of the dataset.

Original Images



Healthy



Shot Hole



Brown Rot

Annotated Images



Healthy



Shot Hole



Brown Rot

3.4 Data Preprocessing

To resizing and normalization the images need to be standardized to a uniform size to ensure compatibility with the deep learning model. Normalization techniques might also be applied to adjust for variations in lighting and color.

Data augmentation is a techniques like image flipping, rotation, and cropping can be used to artificially expand the dataset and improve model generalizability.

Data cleaning is removing blurry, low-quality images or those with mismatched labels is crucial for maintaining data integrity.

3.5 Data Acquisition

The process of acquiring a comprehensive dataset of apricot fruit images necessitates careful planning and execution. This involves capturing images under controlled conditions to ensure optimal lighting and background consistency, as well as collaborating with farmers to obtain images that reflect the diverse environmental variations encountered in real-world settings.

3.6 Block diagram and explanation

This Figure 3.3 block diagram outlines the systematic workflow for building a deep learning

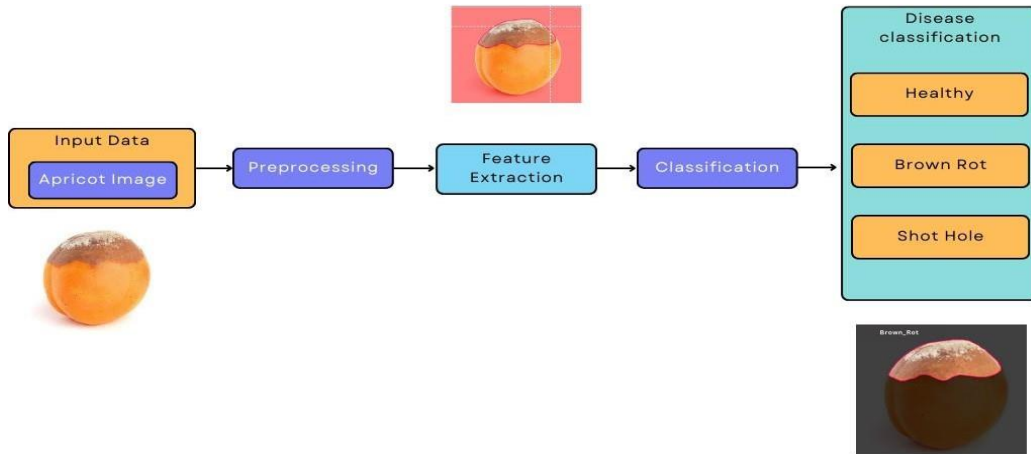


Figure 3.4 Block diagram for apricot fruit disease detection system

model for multi-disease detection in apricot fruit. Each block represents a crucial step, from data acquisition and preprocessing to model training, testing, and ultimately, disease prediction on new images. By following this approach, we can enhance the power of deep learning to achieve accurate and automated disease detection in apricot orchards.

3.7 Deep Learning Model

After being created and trained, the deep learning model's unique architecture was adjusted to account for variables including the quantity of the dataset and the intricacy of the disease. Preprocessed photos are fed into the model, which learns to classify them into several categories, such as healthy and particular illness states. Feature extraction is an essential part of our proposed deep learning-based approach for multi-disease identification in apricot fruit. In order for the deep learning model to effectively use the raw image data for disease classification, it must first be transformed into a more condensed and informative set of features.

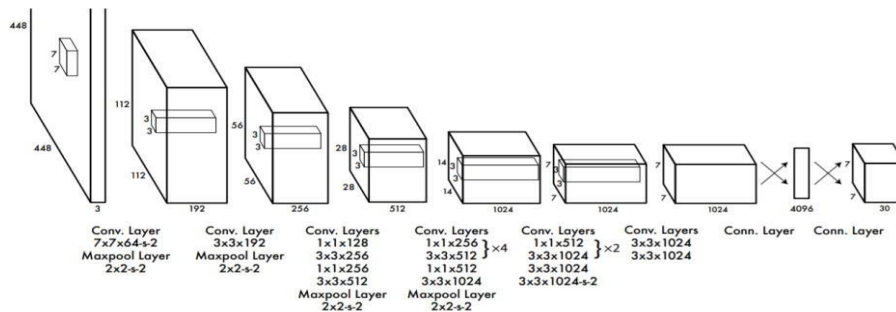
In this context, there are two main methods for extracting features: deep learning-based extraction and manual feature extraction. Using domain knowledge, manual feature extraction entails creating features that are thought to be pertinent for illness identification. These features could include color features (i.e., statistical measures of color distribution across various fruit regions, such as mean and standard deviation); texture features (i.e., image texture analysis captures surface smoothness or roughness); and shape features (i.e., geometric properties such as circularity, aspect ratio, or lesion presence). While manual feature extraction offers some control over the features that are used, it is frequently laborious, necessitates a high level of domain expertise, and may miss subtle variations that could be signs of a disease.

In contrast, deep learning-based feature extraction leverages the power of deep learning models to automatically learn and extract informative features directly from the image data. This approach not only saves time but also allows the

model to capture a broader range of subtle variations that might be missed through manual feature design, leading to more accurate disease classification.

3.8 Yolo Architecture

You Only Look Once (YOLO) M. E. ATİK et al, [13] is an open-source object detection algorithm based on convolutional neural networks. YOLO is among the most well-known deep learning algorithms, and it stands out with its speed thanks to its single-stage detection architecture. Figure 3.5 explain the architecture of yolomodel.



Before YOLO traditional object detection systems relied on separate components: classifiers to identify objects and localizers to pinpoint their location in the image. These systems would scan the image multiple times at different scales and locations, passing each section through the classifier. Areas with high classification scores (likely containing objects) were then identified as potential objects.

YOLO's Different Approach, YOLO (You Only Look Once) revolutionized object detection by using a single neural network for the entire process. Instead of separate components, YOLO treats object detection as a single regression problem. The network divides the image into a grid of regions. For each region, it predicts two things:

- **Bounding boxes:** These boxes indicate the potential location and size of an object within that region.
- **Class probabilities:** The network estimates the likelihood of each region containing a specific object class.
- Finally, these bounding boxes are weighted based on the predicted class probabilities, giving more importance to detections with a higher chance of being a true object.
- **YOLO's Grid System and Object Detection,** YOLO breaks down the input image into a grid of squares. The size of this grid can vary depending on the specific YOLO version (e.g., 3x3, 5x5, or 19x19).
- Each grid cell acts like a mini-detector, responsible for two key tasks:
- **Object Presence:** It predicts whether an object exists anywhere within its

- **Object Attributes:** If an object is present, the cell predicts its attributes like:
Center Point: The x and y coordinates of the object's center point within the image.
- **Size:** The width and height of the object.
- **Class:** The probability of the object belonging to a specific class.

Based on these predictions, YOLO creates bounding boxes around the detected objects. These boxes indicate the location and size of the objects in the image.

Combining Information: The Prediction Vector:

YOLO uses a concept called a "prediction vector" for each grid cell. This vector combines all the information the cell predicts:

Confidence Score: This score indicates how certain the model is about the object's presence and its attributes.

Bounding Box Coordinates: The x and y coordinates of the center point, along with the width and height, define the bounding box.

Class Probability: The probability of the object belonging to a specific class. By analyzing these prediction vectors from all grid cells, YOLO achieves efficient object detection within the entire image

3.9 YOLOv8

The architecture of YOLOv8 is meticulously designed with distinct components to optimize object recognition, balancing precision and speed. The model is structured into three primary segments: the backbone, the neck, and the head, each playing a critical role in the model's overall functionality.

3.9.1 Backbone

As the fundamental part, the backbone is in charge of extracting important features from the input image by means of a sequence of layered procedures that advance these features from simple edges and textures to intricate object representations. Convolutional Layers are the first in this process; they identify basic patterns like edges and advance to more complex features as input flows through the network. Cross Stage Partial (CSP) Blocks are incorporated into the backbone to improve learning efficiency. They split and merge feature maps, which successfully lowers computational costs and memory use without compromising accuracy. Residual Blocks also solve the vanishing gradient issue, making it easier to build deeper networks by combining the output of one layer with the input of the following. And lastly, the Pyramid of Space

3.9.2 Neck

The neck architecture is critical to the integration and refinement of the characteristics that the backbone extracts, allowing feature pyramids to be created, which are necessary for multi-scale object detection. Path Aggregation Network (PANet) greatly improves this process by strengthening the feature hierarchy through the efficient combination of low-level and high-level features,

which increases the precision of object localization and detection. Concurrently, multi-scale feature maps are produced by Feature Pyramid Networks (FPN), which are essential for guaranteeing that the model correctly identifies objects with varied sizes by combining information from several layers.

3.9.3 Head

The part of YOLOv8 that is in charge of carrying out the actual detecting procedure is the head. It predicts bounding boxes, class probabilities, and objectness scores using the finely tuned characteristics that the neck provides. The Detect Layer, which creates bounding boxes with exact details for each grid cell, including centre coordinates, width, height, objectness score, and class probabilities, is a crucial component of this step. Anchor boxes are used by YOLOv8. The model predicts offsets with respect to these anchors to adjust the bounding box coordinates and improve alignment with the objects that are detected. Higher scores indicate greater confidence in the likelihood that an object exists within a given bounding box. The objectness score quantifies this likelihood. Furthermore, the model calculates class probabilities for each bounding box, which allows it to accurately identify the class of the detected object.

3.10 Training

When combined with data augmentation, the YOLOv8 model outperforms current state-of-the-art detection techniques in the multi-disease detection of Apricot fruit. However, the requirement for hyperparameters customized for every individual dataset means that achieving optimal performance with deep learning models is still difficult.

The multi-disease detection of Apricot fruit was accomplished using the YOLOv8 model, yielding results that, when the input dataset for this study was first resized, then split into three sets: 70% training, 20% validation, and 10% test. Initially, the dataset was developed, and the images were manually annotated that is a vital phase when a human expert indicates the location and class of objects within each image. This manual annotation is essential for training object detection algorithms as it provides the ground truth data needed for the algorithm to learn effectively. Finally, an analysis is conducted, achieving state-of-the-art results with YOLOv8. Extensive experiments were carried out using Stochastic Gradient Descent with Momentum (SGDM) and various batch sizes to optimize the model's

Stochastic Gradient Descent with Momentum (SGDM) is employed to minimize the loss function in neural networks. The algorithm updates the model's bias and weights, which are hyperparameters, by making incremental adjustments in the direction of the negative loss gradient. In SGDM, each step towards minimizing the loss function is conducted using mini batches. The size of these mini-batches, referred to as the Mini-Batch Size, is crucial as it influences the evaluation and updating of hyperparameters and the computation of the loss gradient. Momentum is added to the process to reduce oscillations, thereby enhancing convergence stability. SGDM typically applies a uniform learning rate across all hyperparameters, which can either be constant or variable. A very small learning

rate may lead to a prolonged training process, while a high learning rate could result in divergence or suboptimal convergence. To avoid an excessively lengthy training process, a maximum number of epochs is often set. An epoch is defined as one complete pass through the entire training dataset. Extensive experiments are typically conducted to determine the optimal number of epochs required for effective training. The Table 3.1 Details of yolov8 hyperparameters show the final results of Yolov8 model.

Parameters	Values
Epochs	150
Optimizer	SGDM
Batch	32
Learning Rate	0.01
Momentum	0.937
Decay	Default

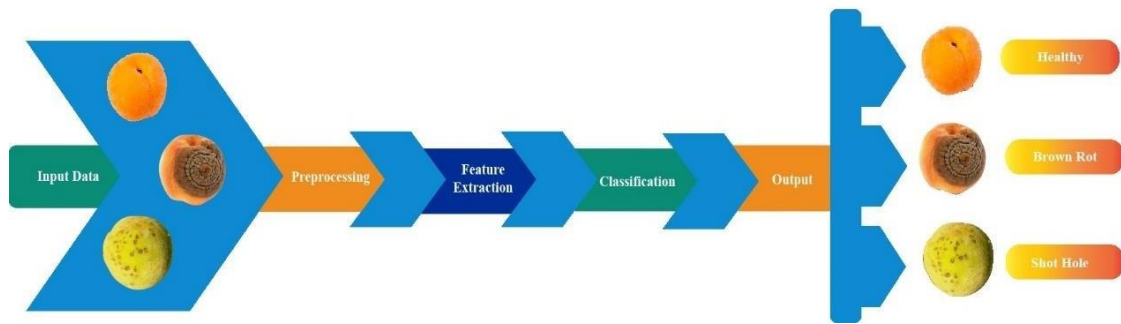
Table 3.1 Details of yolov8 hyperparameters

3.11 Hardware specifications

Google Colab for GPU is used to implement on Microsoft windows 11. Experiments are conducted on intel(R) core i5 with CPU rate 2.60 GHz with installed RAM 16.0 GB.

3.12 Disease prediction

Once trained and evaluated, new apricot fruit images can be input into the model. The model predicts the presence or absence of specific diseases based on the learned patterns from the training data. Figure 3.6 Workflow of apricot disease detection explains the process of prediction.



3.13 Disease prediction via Yolov8

This figure is showing that how our model yolov8 predicted the diseases after the completion of the entire training process. This Figure 3.7 yolov8 predicting sample as healthy is showing that the sample is under inspection is healthy apricot with 89% accuracy.



This Figure 3.8 Yolov8 predicting sample as Infected predicting sample image as infected images with a disease named as brown rot which generally rotten our model is predicted this image as brown rot with 92% accuracy.



This Figure 3.9 Yolov8 predicting sample as infected is showing that the sample under inspection is a diseased image named as shot hole which is good prediction by model with 91% accuracy.



3.14 Conclusion

This literature review highlights the significant advancements made in the fields of plant disease detection, classification, and segmentation using machine learning and deep learning techniques. Research has demonstrated the promising potential of several models, including CNN, EfficientNet, and U-net, in accurately classifying fruit illnesses. There are still, however, issues with model transparency, data imbalance, and dataset availability, especially when it comes to the identification of apricot disease. The field could be further advanced by addressing these issues with larger datasets, better classification methods, and increased explainability. By filling in these gaps, our research will aid in the creation of scalable, accurate, and efficient multi-disease detection algorithms for apricot fruits. Significant improvements in agricultural practices and food security will result from this effort, particularly in apricot-producing regions such as Gilgit-Baltistan.

Chapter 4

IMPLEMENTATION

CHAPTER 4:

4. Implementation

We have carefully selected high-quality data from a range of sources, focusing on the precise identification and classification of diseases, to enable the efficient and accurate diagnosis of apricot diseases. We have put into practice a strong strategy built around a cutting-edge deep learning model to meet these goals. In particular, we have implemented the YOLO (You Only Look Once) architecture, which is well-known for its remarkable effectiveness and performance in object detection applications. The YOLO model is particularly well-suited for the challenging task of identifying and classifying various diseases in apricot fruits because of its quick and accurate detection of multiple objects within an image.

Our research and development process has been meticulously divided into the following key stages:

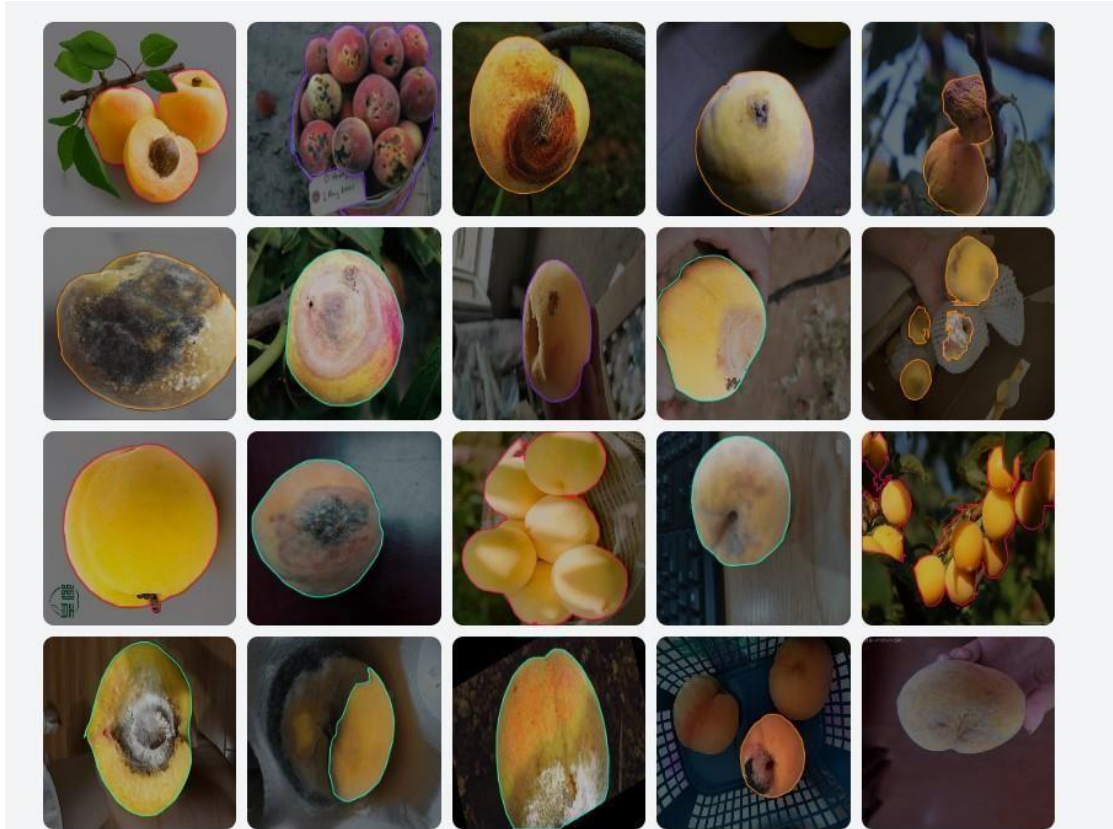
- Data Collection is gathering a comprehensive dataset of apricot fruit images, encompassing both healthy and diseased specimens.
- Implementing the YOLO architecture for disease detection, leveraging its powerful capabilities.
- Procedures is to establish detailed procedures for data preprocessing, model training, and evaluation.
- Performance Metrics of selecting appropriate performance metrics to assess the model's accuracy, precision, recall, and F1-score.
- Results and Discussion is to analyze the model's performance, interpreting the results, and discussing the implications.
- Graphs to visualize the results through informative graphs to enhance understanding and interpretation."

This revised text provides a clearer and more concise explanation of the research process, highlighting the key steps involved and the use of a cutting-edge deep learning model.

4.1 Data Collection

In the process of image acquisition, obtaining high-resolution images of apricot fruit is crucial. This may involve capturing images in orchards under controlled conditions or collaborating with farmers to gather images from diverse growing environments, ensuring a comprehensive dataset. It is important that the dataset includes a wide representation of apricot diseases, covering various diseases at different stages of development to provide the model with the necessary variability for accurate classification. Additionally, a significant portion of the dataset should consist of images of healthy apricots, establishing a clear baseline for distinguishing between healthy and diseased fruit.

Each image in the dataset needs to be meticulously labeled, indicating the presence or absence of specific diseases. This labeling process can be conducted manually by experts or through a combination of manual and automated techniques to ensure accuracy. Figure 4.1 **Error! Reference source not found.** illustrates a sample of the data collected for our dataset, highlighting the diversity and quality of the



images used in the study.

4.2 Procedures

- Step 1: browse to take picture from the internal storage and upload the picture
- Step 2: identify the pattern of the image whether it is infected or healthy.
- Step 3: The model will determine the nature of image
- Step 4: The input image is predicted as infected or as a healthy image with accuracy parameter

4.3 Performance Matrix

The performance matrix is some we can call them like accuracy, precision and F1-score and many more. The matrix to evaluate the performance of our model is precision, recall, mAP(mean average precision) and confusion matrix these are the typical matrix for our model.

4.3.1 Precision

The precision metric is used to overcome the limitation of Accuracy. The precision determines the proportion of positive prediction that was actually correct. It can be

Figure 4.1 Samples of data

calculated as the True Positive or predictions that are actually true to the total positive predictions (True Positive and False Positive).

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

4.3.2 Recall or Sensitivity

It is also similar to the Precision metric; however, it aims to calculate the proportion of actual positive that was identified incorrectly. It can be calculated as True Positive or predictions that are actually true to the total number of positives, either correctly predicted as positive or incorrectly predicted as negative (true Positive and false negative).

The formula for calculating Recall is given below

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FN})$$

4.3.3 Mean Average Precision (mAP)

mAP is a widely used metric in object detection and information retrieval tasks. It provides a comprehensive measure of a model's performance by considering both precision and recall. mAP calculates the average precision for each class in a dataset and then takes the mean of these average precision values. This provides a single, overall metric that captures the model's ability to correctly identify objects of interest while minimizing false positives and false negatives.

$$AP = \sum (\text{precision}(k) * (\text{recall}(k) - \text{recall}(k - 1)))$$

Where:

- k is the rank of a positive instance.
- precision(k) is the precision at rank k.
- recall(k) is the recall at rank k.

$$mAP = (AP1 + AP2 + \dots + APN) / N$$

Where:

- N is the total number of classes.

4.3.4 Confusion Matrix

A confusion matrix is a tabular representation of prediction outcomes of any binary classifier, which is used to describe the performance of the classification model on a set of test data when true values are known. The confusion matrix is simple to implement, but the terminologies used in this matrix might be confusing for beginners. A typical confusion matrix for a binary classifier looks like the below image (However, it can be extended to use for classifiers with more than two classes). The Figure 4.2 show the confusion matrix.

In general, the table is divided into four terminologies, which are as follows

- True Positive is interpretation we predicted positive and it's true.
- True Negative is interpretation we predicted negative and it's true.
- False Positive (Type 1 Error) is interpretation we predicted positive and it's false.
- False Negative is interpretation we predicted negative and it's false.

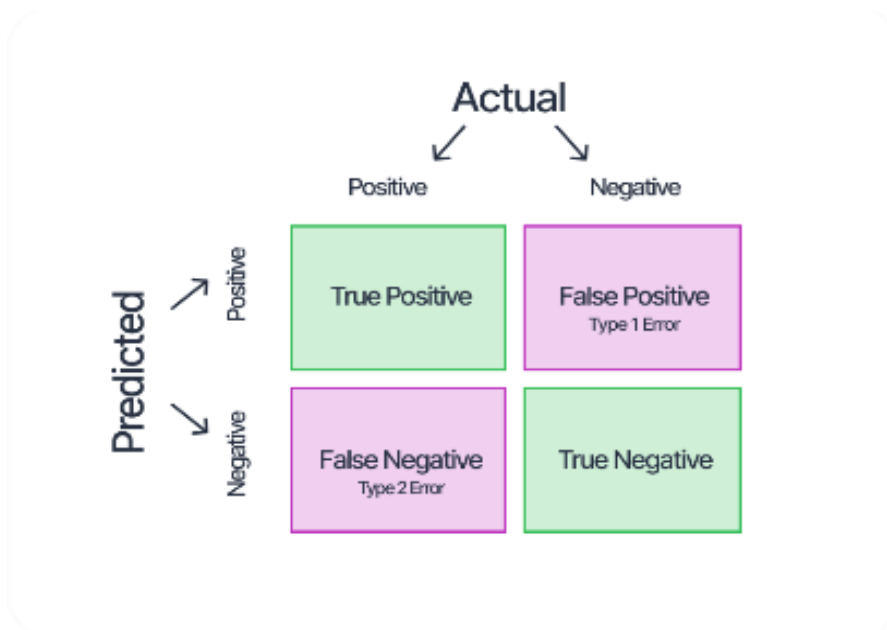


Figure 4.2 Confusion Matrix

4.4 Results and Discussion

This is what we have done a comparative study among the models given below in the table with some information of epochs, batch size, optimizer and the efficiency of each model respectively.

Yolov8 is the model through which we have got most satisfying results. It is with 150 epochs, 32 batch size, optimizer we used is SGD and we get results in terms of recall, precision and mAp 92.2%, 92.5% and 95% respectively. In our comparative research with other models briefly discussed below, yolov8 is one of the best performing models with our dataset through which we have made an integration with web application in the light of the performance of the model yolov8.

CNN stands for convolutional neural network which is mainly used for classification tasks and here we have CNN with our dataset for comparative study and we have got better results with minimal epochs although our motive is not classification still, we have used for the comparison and we have the accuracy of 91.83% with 10 epochs, batch size 32 and optimizer Adam.

Vgg16 is also used for classification, but we have tried in our comparative research as compared to CNN, the performance of Vgg16 is a little bit low and we have got results in terms of accuracy 89.42% with 10 epochs, batch size 32 and optimizer Adam.

Yolo v9 is the recent version of the yolo models and the performance of the yolo v9 is not good enough with our dataset. Yolo v9 required a lot of resource like GPU and machine specification to train the model with large size dataset. We have got better results with minimal epochs in terms of recall, precision and mAp 77.0%, 76.0% and 83% respectively with 100 epochs, batch size 16 and optimizer

Adam.

In case of YOLOv5 the computational usage is not bad YOLOv5 used less computational resources than YOLOv8 and YOLOv9. However, it is using less resources and giving results comparatively low than other two models YOLOv8 and YOLOv9. Through this model we have got result with the minimal epochs we have got better result than with minimal epochs in terms of recall, precision and mAP 90.0%, 90.0% and 93% respectively with 100 epochs, batch size 16 and optimizer Adam.

Now at last we have YOLO Nas with least performance through this model we have got result with the minimal epochs we have got better result than with minimal epochs in terms of recall, precision and mAP 93.7%, 91.7% and 95.6% respectively with 300 epochs, batch size 16 and optimizer Adam. It takes a lot of time and resources and gives minimal results.

In our comparative study we have completely addressed the best efficiency of the listed models shown in Table 4.1.

Model	Epochs	Batch	Optimizer	Accuracy	Recall	Precision	mAP
YOLOv8	150	32	SGD	-	92.2%	92.5%	95%
CNN	10	32	Adam	91.83%	-	-	-
Vgg16	30	32	Adam	89.42%	-	-	-
YOLOv9	100	16	Adam	-	77%	76%,	83%
YOLOv5	100	32	Adam	-	90%,	90%,	93%
YOLO Nas	300	64	Adam	-	93.7%,	91.7%	95.6%

Table 4.1 Result and discussion

4.5 State-of-the-art comparison

This table of comparison is going to explain a comparative aspect of the listed models. These models are being used for detection and as well as classification of the data. In our case, we have chosen down some of the models in order to make a comparison while considering YOLOv8 especially for the detection of the disease in apricot fruits. However, the comparing models in this research are considering a single disease and the classification of varieties and in our case, we are addressing multiple disease detection from our dataset using YOLOv8.

Detection						
Reference	Models	Precision	F1-score	Accuracy	Recall	mAP
[6]	Alex-Net	96.3%	-	97.354%	96.5%	-
	Google-Net	96.989%	-	97.7%	96.8%	-
	SVM	95.4%	-	92.9%	92.5%	-
	Random forest	94.6%	-	93.1%	93.8%	-

[12]	CNN	-	-	98.6%	-	-
Classification						
Reference	Models	Precision	F1-score	Accuracy	Recall	mAP
[3]	CNN	-	-	100%	-	-
[5]	CNN	-	-	98.3%	-	-
[7]	AlexNet	-	-	-	-	-
	KNN	-	-	-	-	-
	Vgg16	-	-	94.8%	-	-
	Vgg19	-	-	-	-	-
Detection via yolov8						
Reference	Models	F1 score	Accuracy	Recall	Precision	mAP
Our Trained Model	Yolov8	92.3%	-	92.2%	92.5%	95%

Table 4.2 State-of-the-art comparison among models

4.6 Graphs

Graphs from the models to visualized the learning of models so here we can visualize the testing and training data as well as lose. The concatenated graphs of the same have been shown below.

4.6.1 Yolo v9

An F1 curve, in the context of object detection models like YOLOv9, is a graphical representation that illustrates the model's performance across different confidence thresholds. It typically plots the F1-score (a harmonic mean of precision and recall) on the y-axis against the confidence threshold on the x-axis.

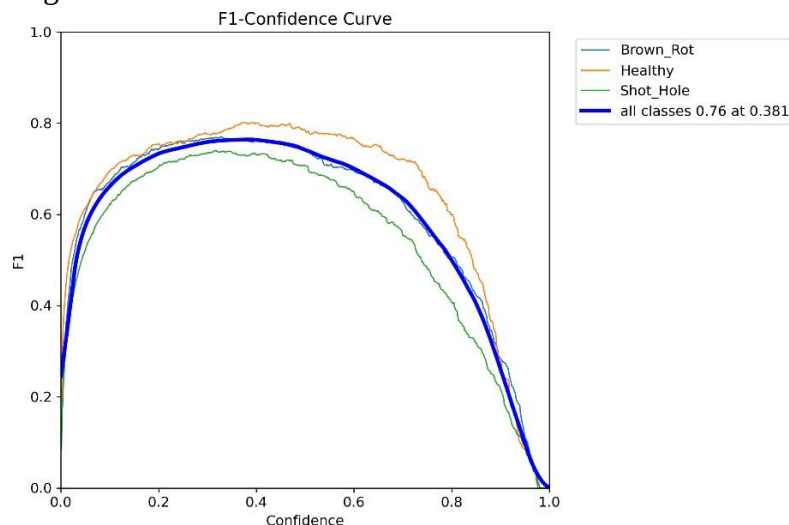


Figure 4.3 F1curve of yolo v9

In the context of object detection, such as with YOLOv9, a confusion matrix is a visualization tool that helps assess the model's performance. It provides a tabular representation of the predicted and actual classes for each detected object.

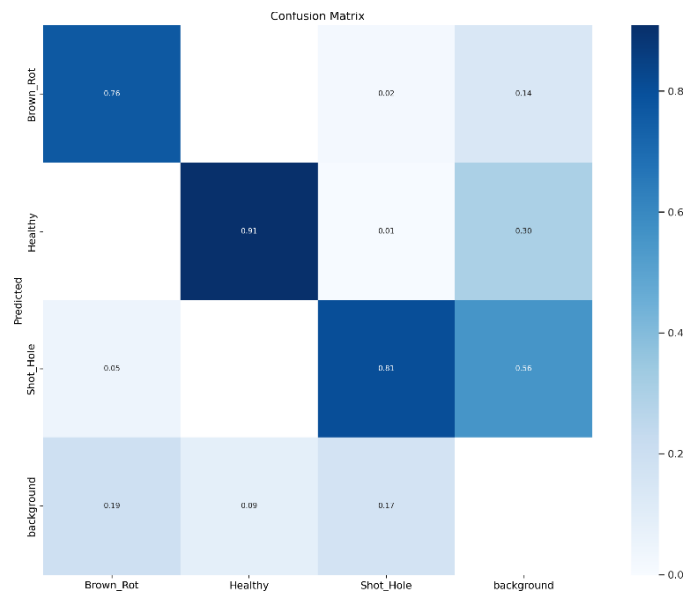
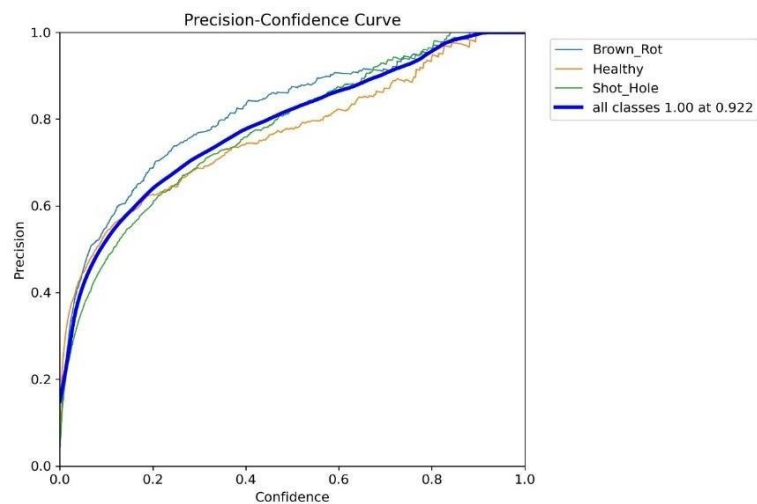


Figure 4.4 Confusion matrix of yolo v9

The precision of model shown in below Figure 4-5



The recall of model shown below Figure 4.6 this is actually expressing the recall parameter of model through which we can assess our false positive predictions of our model and through this graph give below simply we can conclude that how the model is getting trained and showing the values in correspondence with epochs.

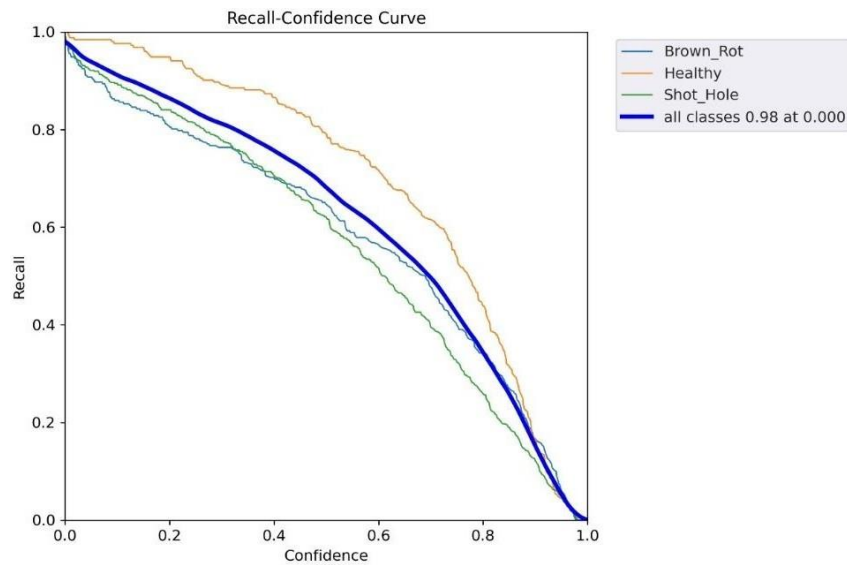
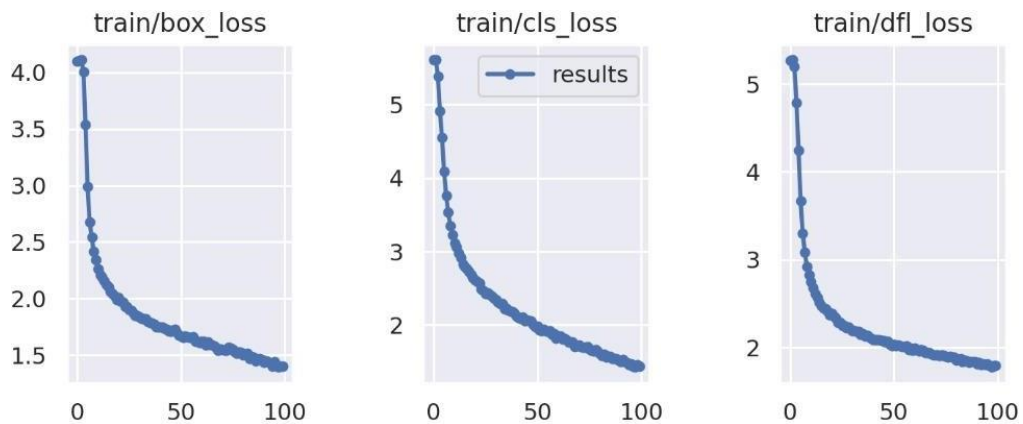


Figure 4.6 Recall curve of yolo v9

Training loss shown in below Figure 4.7 in YOLOv9 is a metric that quantifies the model's performance during the training process. It measures the discrepancy between the model's predicted output and the ground truth labels.

Training loss in YOLOv9 is a quantitative metric that evaluates the model's performance throughout the training process. It measures the disparity between the model's predicted outputs and the corresponding ground truth labels. In essence, training loss serves as a valuable indicator of the model's ability to learn and accurately predict the correct outcomes.



4.6.2 Yolo v8

Through confusion we can visualized that how the prediction is true and how much is incorrect. Figure 4.8 shows the confusion matrix of yolov08. A confusion matrix is a fundamental tool in classification problems used to evaluate the performance of a classification algorithm. It summarizes the results of a classification task by comparing the predicted labels with the true labels.

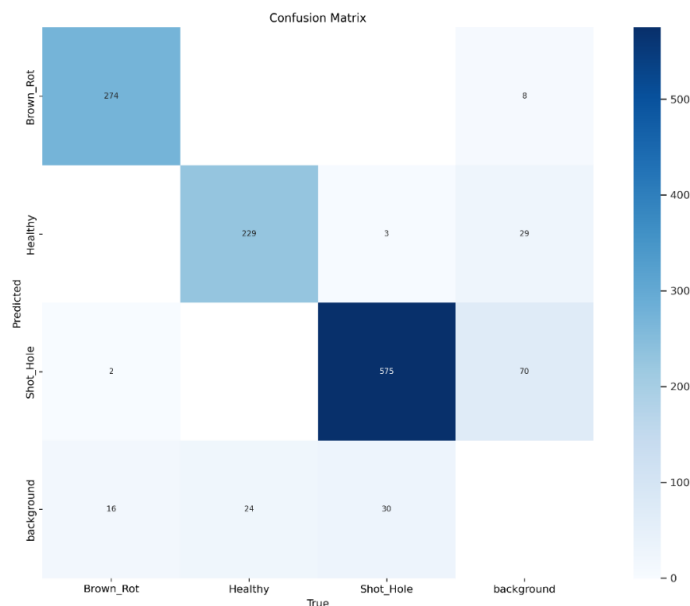
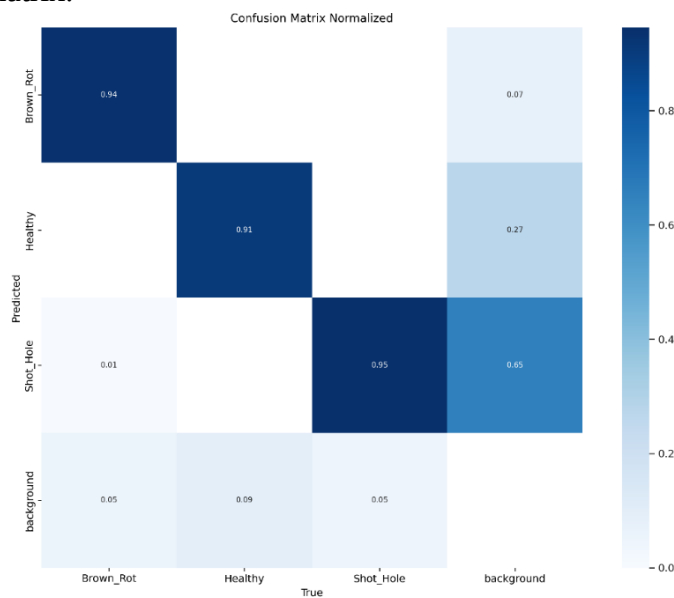
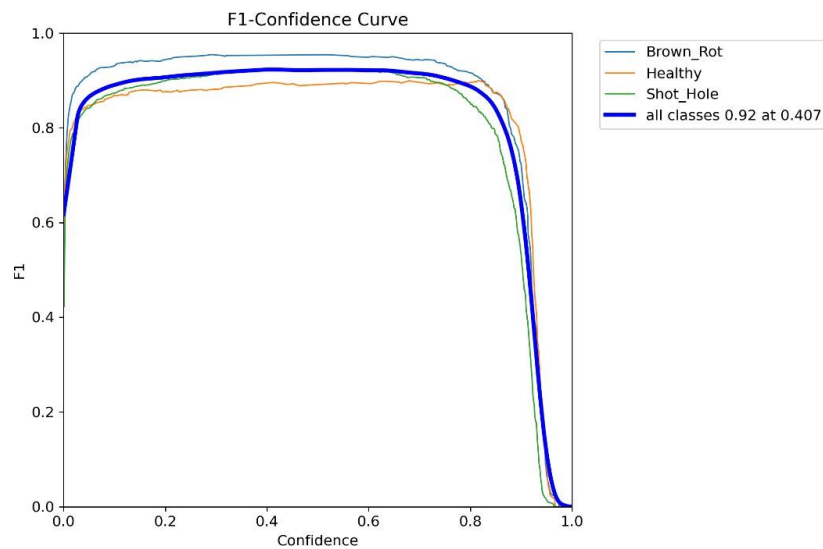


Figure 4.8 Confusion Matrix YOLOv8

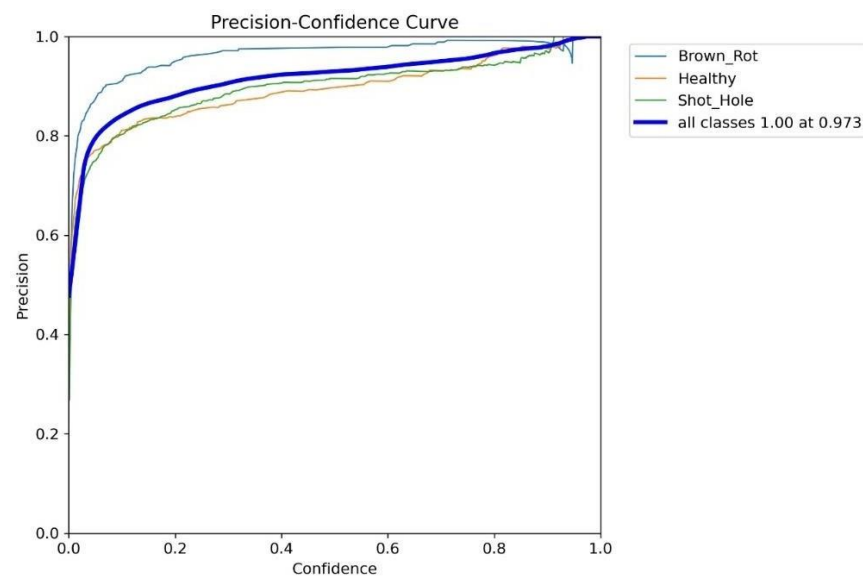
In a normalized confusion matrix, the counts in each cell of the matrix are scaled by the total number of instances in the corresponding true class. This normalization makes it easier to compare the performance across different classes, especially when class distributions are imbalanced. Figure 4.9 shows the normalized confusion matrix.



A confidence curve in this context might refer to a graphical representation of how the confidence of the model predictions varies across different F1 scores or thresholds. For example, you might plot the F1 score against various thresholds to see how the model's performance changes with different levels of confidence. Figure 4.10 shows the F1 curve of Yolov08.



The precision curve of model shown in below Figure 4.11



The recall curve of model shown in Figure 4.12

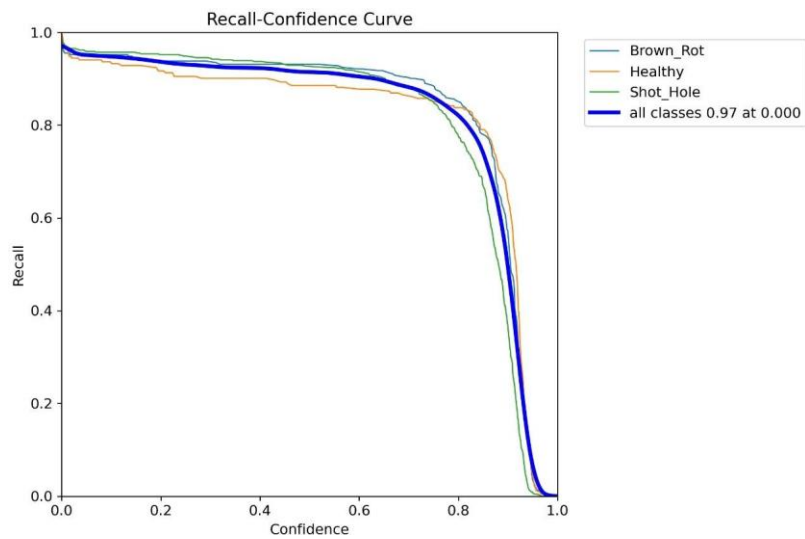
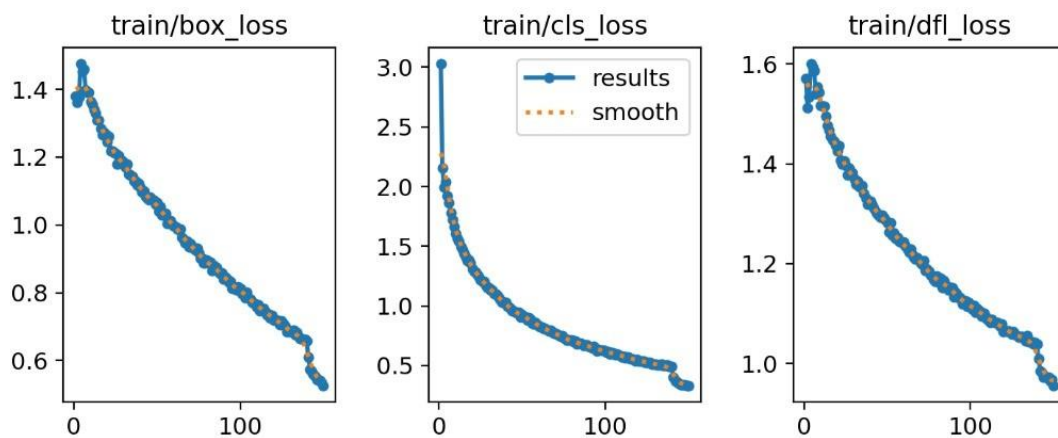


Figure 4.12 Recall of yolo v8

Training loss in YOLO shown in Figure 4.13 is a metric that quantifies the model's performance during the training process. It measures the discrepancy between the model's predicted output and the ground truth labels.

Training loss in YOLO is a quantitative metric that evaluates the model's performance throughout the training process. It measures the disparity between the model's predicted outputs and the corresponding ground truth labels. In essence, training loss serves as a valuable indicator of the model's ability to learn and accurately predict the correct outcomes.



4.6.1 Yolo v5

A confusion matrix is a useful tool for evaluating the performance of a classification model, including object detection models like YOLOv5. It provides a detailed breakdown of how well the model's predictions match the true labels. For object detection, a confusion matrix can be adapted to reflect the model's performance on individual classes.

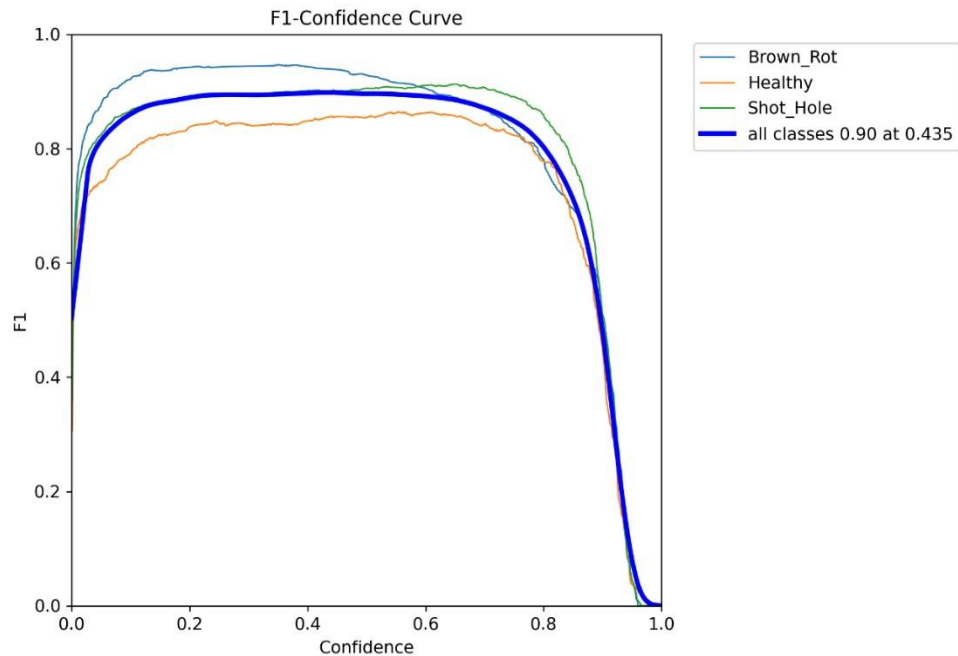
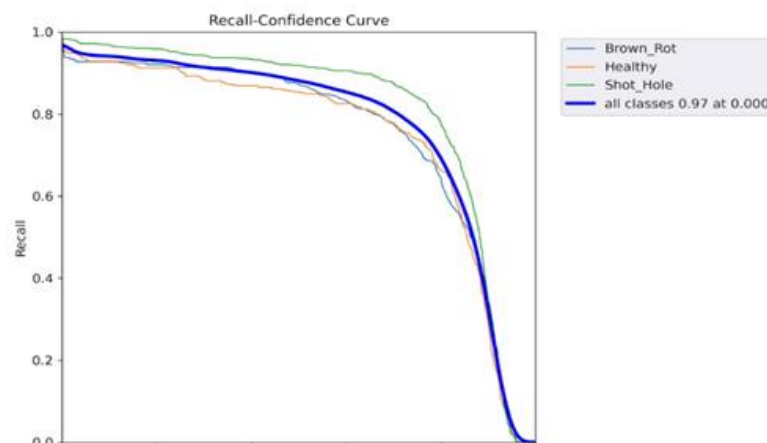
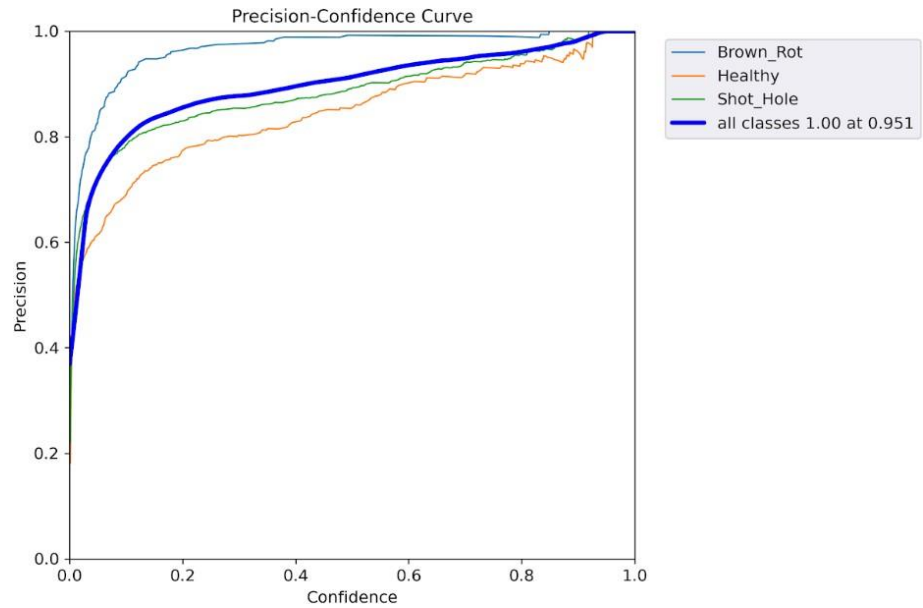


Figure 4.14 F1-Confidence Curve YOLOv5

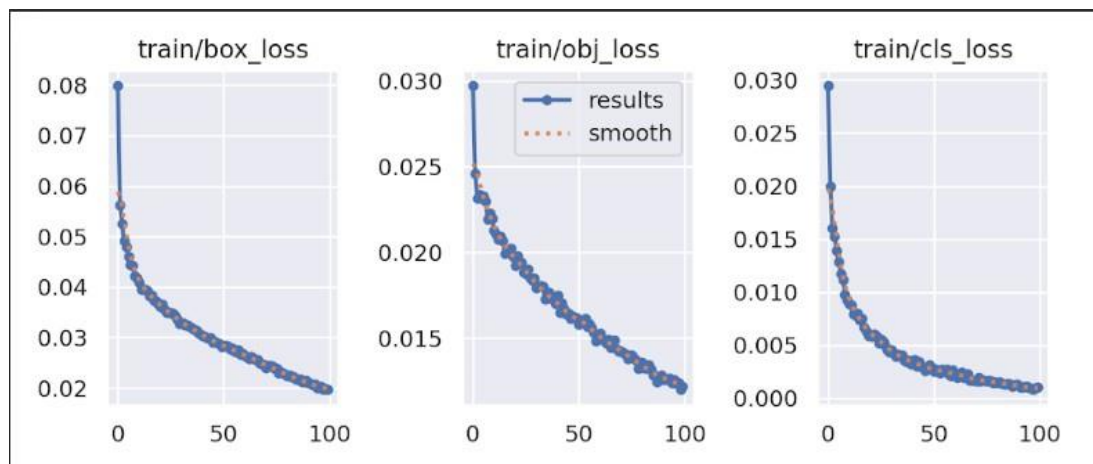
The F1 confidence curve shown in Figure 4-15 is a visualization used to evaluate the performance of classification models, particularly in tasks involving object detection or other scenarios where precision and recall are critical. It helps you understand how the F1 score (a harmonic mean of precision and recall) changes across different confidence thresholds. The Recall of model shown in Figure 4.15



The precision curve of model shown in Figure 4.16



A common visualization technique is to plot the training loss over epochs. This can help identify trends, plateaus, and potential issues in the training process. Figure 4.17 shows the training loss of the model YOLOv5.



4.6.2 CNN

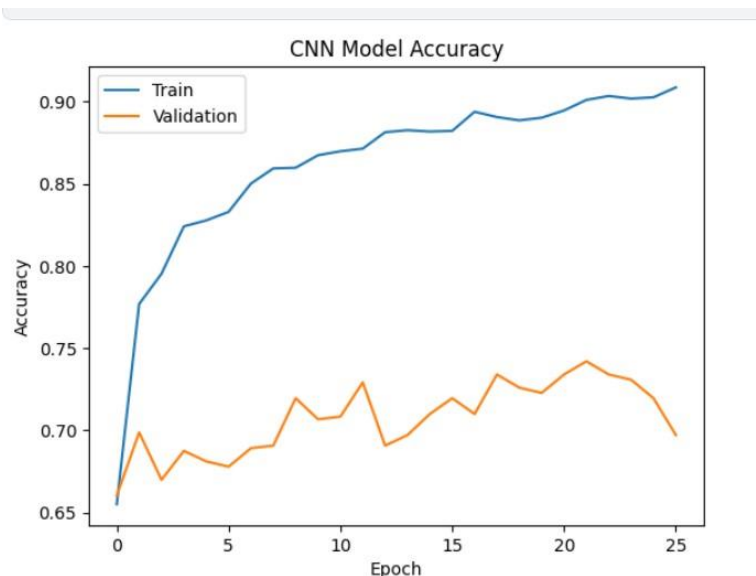
Evaluating the accuracy of a Convolutional Neural Network (CNN) model depends on various factors, including the specific dataset, the problem being solved, and the model's architecture

The accuracy of a CNN model is heavily influenced by the dataset it is trained on. For example, a model trained on the MNIST dataset (handwritten digits) might achieve accuracy over 98%, while a model trained on a more complex dataset like ImageNet (which contains millions of diverse images) might have lower accuracy but still be considered highly effective if it performs well.

Accuracy is just one metric shown in Figure 4.18. Depending on the application, other metrics like precision, recall, F1-score, and area under the curve (AUC) might be more relevant. For classification tasks, accuracy is often used, but for tasks like object detection or segmentation, you might look at metrics like Intersection over Union (IoU) or mean Average Precision (mAP).

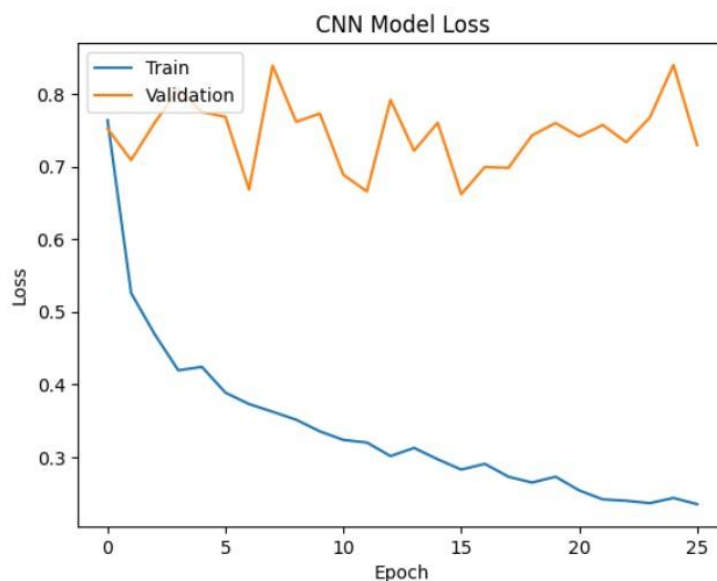
To ensure that a model is not overfitting or underfitting, you should evaluate it on both training and validation datasets. Accuracy on the training set might be high, but if it is significantly lower on the validation set, this could indicate overfitting.

Using techniques like k-fold cross-validation can provide a more robust estimate of a model's accuracy by training and testing it on different subsets of the data.



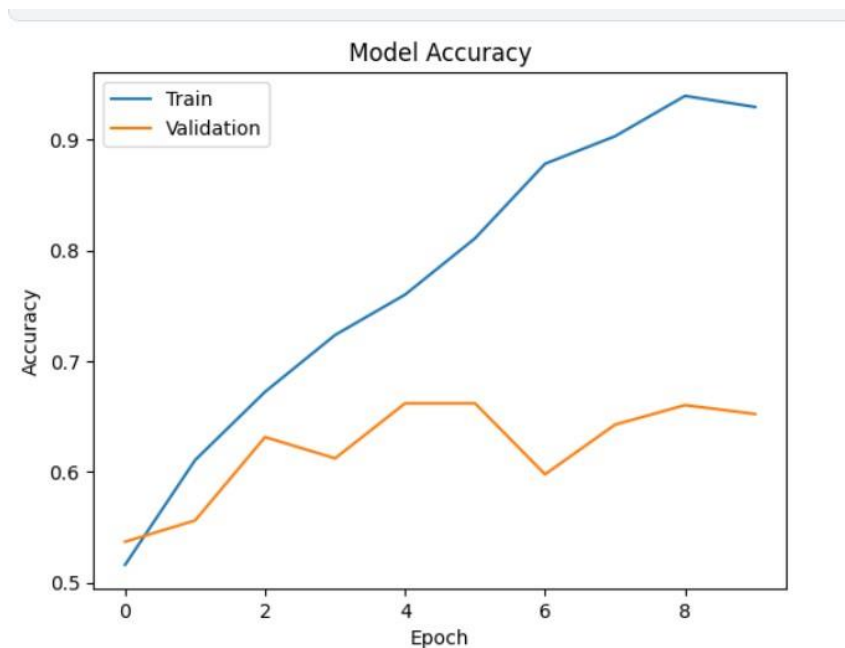
In Convolutional Neural Networks (CNNs), loss is a measure of how well the model's predictions match the actual target values. It's a critical metric for evaluating and training the model. The loss function you use depends on the type of problem you're solving (e.g., classification, regression, etc.). **Error! Reference source not found.** emphasis the CNN model loss.

In the realm of Convolutional Neural Networks (CNNs), loss serves as a quantitative metric that measures the degree of discrepancy between the model's predictions and the actual target values. This critical metric plays a pivotal role in evaluating and refining the model's performance. The specific loss function employed is contingent upon the nature of the problem being addressed, such as classification, regression, or other relevant task



4.6.1 Vgg16

The VGG16 model is a convolutional neural network (CNN) architecture introduced by the Visual Geometry Group (VGG) from Oxford University. It's known for its deep architecture and uniform design, using 16 weight layers (13 convolutional layers and 3 fully connected layers). Figure 4.20 shown the VGG model accuracy on our dataset.



For Classification: Typically, VGG16 uses categorical cross-entropy loss for classification tasks. This loss function is appropriate when the output is a probability distribution across multiple classes.

Measures the difference between the true class labels and the predicted probability distribution. If adapted for binary classification, binary cross-entropy might be used instead. The loss computed on the training dataset. This should decrease as the model learns from the training data. The loss computed on a separate validation dataset. This helps to monitor the model's generalization and check for overfitting. Figure 4.21 shows the VGG Model Loss.

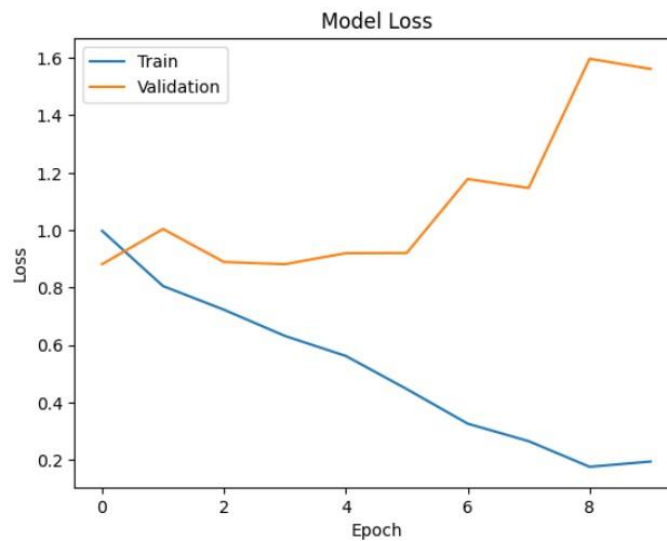
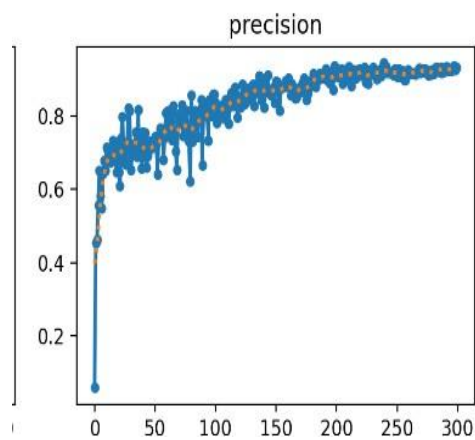


Figure 4.21 VGG Model Loss

4.6.2 Yolo Nas

YOLO-NAS (You Only Look Once - Neural Architecture Search) is a version of the YOLO (You Only Look Once) object



detection framework enhanced by neuralarchitecture search techniques. YOLO models, including YOLO-NAS, are widely used for real-time object detection due to their balance of speed and accuracy. When evaluating YOLO-NAS models, key metrics include Precision, Recall, and Mean Average Precision (mAP). Here's how these metrics are defined and applied. **Figure 4.22 and**

Figure 4.23 show the accuracy and recall of YoloNAS.

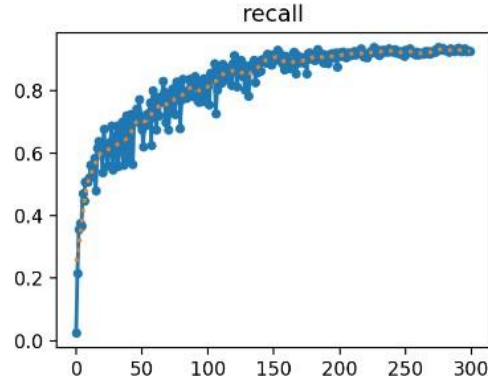


Figure 4.23 YoloNAS Recall

YOLO NAS automatically searches for the best neural architecture based on performance metrics like training loss shown in Figure 4.24. This means the loss function can be more complex as it's used to guide the search process.

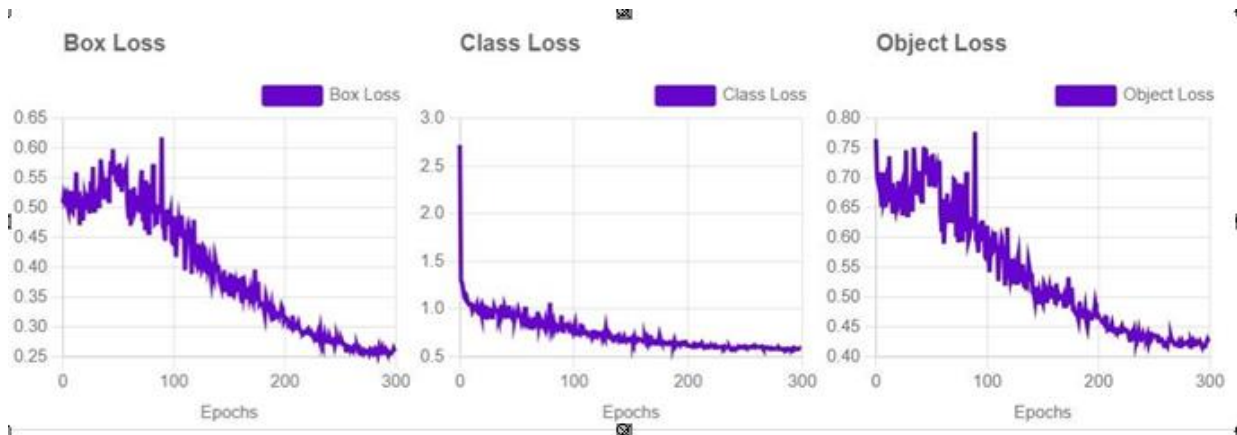


Figure 4.24 Training Loss

4.7 Conclusion

In this study, we implemented a variety of deep learning models to detect multiple diseases in apricot fruits, with a primary focus on YOLOv8 due to its outstanding performance in object detection tasks. After training with 150 epochs and a batch size of 32, YOLOv8 proved to be the most successful model through comprehensive testing and evaluation, obtaining high precision (92.5%), recall (92.2%), and mean average precision (95%). On the other hand, several models like CNN, YOLOv9, VGG16, and YOLO NAS were also evaluated for comparison. Although these models produced acceptable results, they fell short of YOLOv8's performance in terms of detection accuracy and computational efficiency. YOLOv8 is the best option for real-world applications because of its exceptional capacity to manage the complexities of various disease detection in real-time, particularly in

agriculture. This study shows how crucial it is to choose the appropriate model for a given job and emphasizes how cutting-edge deep learning approaches can enhance disease detection and classification in fruits such as apricots.

Chapter 5

Conclusion

CHAPTER 5:

5. Conclusion

In this study, deep neural networks are employed for the classification and detection of Apricot diseases, with a comparative analysis conducted to assess their effectiveness. To prevent overfitting, data augmentation techniques are utilized. Various experiments are performed using different hyper-parameter settings tailored to the proposed approaches. The classification performance of pre-trained models, including CNN and VGG-16, is first evaluated by extracting learned features, with different network layers being replaced to achieve optimal results. Subsequently, the detection performance of pre-trained models YOLOv05, YOLOv08, and YOLOv09 is assessed by extracting learned features, aiming to determine the configurations that yield the best performance.

In the study, two different mechanisms have been used for evaluation i.e., classification and detection separately has been done. For the classification task, Convolutional Neural Networks (CNN) and the VGG-16 architecture were utilized. The experimental results demonstrated that the features extracted by these models were highly effective, leading to optimal performance in classification tasks. Furthermore, the incorporation of data augmentation techniques significantly enhanced the model's ability to recognize and classify the data. This improvement was not merely incremental, it led to a substantial boost in both accuracy and robustness, allowing the model to generalize more effectively on unseen data. The data augmentation process played a crucial role in reducing overfitting, ensuring that the model maintained high performance across various datasets. Experimental results have shown that the proposed approach using the model with data augmentation resulted in better recognition/classification performance and outperformed the state-of-the-art research.

For the detection task, the proposed method achieved impressive performance metrics, including a Precision of 92.5%, Recall of 92.2%, and a mean Average Precision (mAP) of 95%. These results underscore the effectiveness of the approach in accurately detecting and classifying diseases. The experimental findings reveal that the proposed YOLO v8 model, enhanced with data augmentation, outperforms current state-of-the-art detection methods. However, optimal performance of deep learning models requires hyperparameters specific to each dataset, which poses a challenge. Additionally, deep learning models demand large volumes of data to outperform other strategies, and training these complex models is exceedingly resource-intensive, necessitating the use of expensive GPUs and numerous workstations. Looking ahead, we plan to further enhance recognition and classification accuracy by exploring alternative feature extraction techniques.

6. Limitation

A significant constraint of the research is the requirement for hyperparameter adjustments unique to every dataset. Precisely adjusting these variables is necessary to achieve optimal performance in deep learning models; yet, this process can be time-consuming and dataset-dependent. Furthermore, in order to beat alternative methods, deep learning models for disease detection and classification require enormous amounts of data. Obtaining such vast datasets is frequently extremely

difficult.

The resource-intensive aspect of training sophisticated models is another drawback. These models can be unaffordable due to their requirement for strong GPUs and several workstations. The enormous computing needs make training and deploying models more challenging

7. Future work

In order to enhance the efficacy of disease classification and detection models, the study intends to investigate alternate feature extraction methods in subsequent research. The pre-defined architectures of CNN and VGG-16, which are frequently utilized in current deep learning models, might not necessarily capture the most pertinent features for certain datasets. The objective is to find strategies that can more accurately capture the underlying patterns in the data by examining various feature extraction techniques, which could result in models that are more reliable and accurate.

Additionally, there is a desire to experiment with more recent neural network topologies or hybrid strategies that incorporate many methods. This could involve utilizing transformer models for enhanced attention processes, group learning, or more sophisticated data augmentation techniques that produce artificial data that closely mimics real-world settings.

Reference

8. Reference

- [1] M. Muhammad *et al.*, "Geostatistical Analysis of Apricot Shot Hole Disease and Influence Factors in District Nagar, Gilgit-Baltistan, Pakistan," *International Journal of Phytopathology*, vol. 11, no. 3, pp. 227-238, 2022.
- [2] C. Chen *et al.*, "Construction of apricot variety search engine based on deep learning," *Horticultural Plant Journal*, vol. 10, no. 2, pp. 387-397, 2024.
- [3] A. Mehmood, M. Ahmad, and Q. M. Ilyas, "On precision agriculture: enhanced automated fruit disease identification and classification using a new ensemble classification method," *Agriculture*, vol. 13, no. 2, p. 500, 2023.
- [4] K. Amari, D. Ruiz, G. Gómez, M. A. Sánchez-Pina, V. Pallás, and J. Egea, "An important new apricot disease in Spain is associated with Hop stunt viroid infection," *European Journal of Plant Pathology*, vol. 118, pp. 173-181, 2007.
- [5] M. H. Saleem, J. Potgieter, and K. M. Arif, "Plant disease detection and classification by deep learning," *Plants*, vol. 8, no. 11, p. 468, 2019.
- [6] M. E. Chowdhury *et al.*, "Automatic and reliable leaf disease detection using deep learning techniques," *AgriEngineering*, vol. 3, no. 2, pp. 294-312, 2021.
- [7] M. TÜRKOĞLU and D. HANBAY, "Apricot disease identification based on attributes obtained from deep learning algorithms," in *2018 International Conference on Artificial Intelligence and Data Processing (IDAP)*, 2018, pp. 1-4: IEEE.
- [8] A. Ahmad, A. El Gamal, and D. Saraswat, "Toward generalization of deep learning-based plant disease identification under controlled and field conditions," *IEEE Access*, vol. 11, pp. 9042-9057, 2023.
- [9] D. Unay and B. Gosselin, "Apple defect segmentation by artificial neural networks," in *Proceedings of BeNeLux Conference on Artificial Intelligence*, 2006: Citeseer.
- [10] V. D. Soni, "Chronic disease detection model using machine learning techniques," *International Journal of Scientific & Technology Research*, vol. 9, no. 9, pp. 262-266, 2020.
- [11] A. Badage, "Crop disease detection using machine learning: Indian agriculture," *Int. Res. J. Eng. Technol*, vol. 5, no. 9, pp. 866-869, 2018.
- [12] U. N. Fulari, R. K. Shastri, and A. N. Fulari, "Leaf disease detection using machine learning," *J. Seybold Rep*, vol. 1533, p. 9211, 2020.